Automated quantitative analysis of mycobacterial cording phenotypes



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Abstract

Automated quantitative analysis of mycobacterial cording phenotypes

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This study presents the development and evaluation of an automated image processing pipeline for the quantitative analysis of Mycobacterium tuberculosis (Mtb) cording phenotypes, addressing a critical need in microbiological diagnostics and research. Mtb, the causative agent of tuberculosis, exhibits clinically relevant cording and non-cording phenotypes, observable under low-magnification microscopy, which are traditionally assessed qualitatively by trained observers, leading to subjectivity. Leveraging a novel dataset of 183 low-magnification images, the pipeline integrates U-Net-based segmentation with Edge-Dilated Weak Masking (EDWM), multi-scale texture feature extraction using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) techniques, and Support Vector Machine (SVM) classification. The segmentation model achieved high class-specific Dice coefficients (0.887 for cording, 0.851 for non-cording) but a lower overall Dice (0.616/0.614) due to multi-class challenges, reflecting strengths in phenotype-specific contexts and limitations in mixed-phenotype generalization. Texture analysis quantified 67 descriptors, with statistical evaluation identifying 24 significant features (e.g., LBP histograms) that distinguish cording (Parental_Phigh) from non-cording (Parental_Pminus) phenotypes, validated by an SVM classifier achieving 94.8% accuracy on 135 unseen samples. Retraining with the 24 features improved accuracy to 95.6%, with perfect recall (1.00) and high average confidence (0.8, 85.9% $\geq$ 0.8). These findings demonstrate the pipeline's efficiency, with potential for real-time clinical applications and contributions to ongoing research on Mtb mechanopathology. Future improvements, including larger, diverse datasets and refined masks, are proposed to enhance overall segmentation and classification efficiency.

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List of Acronyms and Glossary

DC	Dice Coefficient
EDWM	Edge Dilated Weak Masking
FDR	False Discovery Rate
IoU	Intersection-over-Union
LBP	Local Binary Pattern
GLCM	Gray Level Co-occurrence Matrix
GPU	Graphics Processing Unit
ML	Machine Learning
MODS	Microscopic Observation Drug-Susceptibility
Mtb	Mycobacterial Tuberculosis
SVM	Support Vector Machine

Chapter 1

Introduction

1.1 Background

The ability to quantitatively analyse microbial cell morphologies and phenotypes is essential for transforming qualitative biological observations into reproducible, data-driven insights. With recent advances in microscopy and imaging techniques, there is demand for effective computational strategies to identify cell biological phenotypes through the quantification of cell phenotype (Ghanegolmohammadi, et al., 2024). In *Mycobacterium tuberculosis* (Mtb), one classic feature is "cording," where multiple bacterial cells form organized, rope-like aggregates under specific growth conditions (Middlebrook, et al., 1947). This phenotype is particularly useful for diagnostic methods such as the Microscopic Observation Drug Susceptibility (MODS) assay, which relies on visual recognition of cords as an indicator of Mtb growth (Caviedes, et al., 2000). The identification of cords in MODS assays is typically performed manually**,** relying heavily on the observer's expertise. This approach can be time-consuming, subjective, and prone to human error, often requiring specialized training and experience to ensure consistent interpretation. The aim of this project is therefore to create an automated quantitative analysis pipeline which will enable objective detection, classification, and measurement of aggregation phenotypes to discriminate between bacterial strains that cord from those that do not. Developing such an image-processing pipeline capable of standardizing data acquisition, segmenting complex bacterial aggregates, and extracting meaningful features to describe these phenotypes would provide a useful framework for comparing phenotypes across strains, conditions, and experimental setups.

1.2 Problem statement

The analysis of mycobacterial cording phenotypes remains predominantly qualitative, relying on visual categorization of microscopy images by trained observers. While this approach has been foundational in diagnostic assays such as the Microscopic Observation Drug Susceptibility (MODS) test, it introduces significant limitations (Jihadah, et al., 2025). Manual assessment is subjective, making it prone to variability between observers, and difficult to scale for high-throughput or comparative studies. Qualitative evaluation restricts the ability to extract nuanced morphological information, such as variations in aggregate organization, size distribution, or structural integrity, which may be critical for understanding strain-specific behaviors and responses to experimental conditions.

To address these challenges, there is a need for an automated, quantitative image processing pipeline capable of accurately identifying, segmenting, and characterizing mycobacterial aggregates across distinct cording phenotypes. Such a system would enable reproducible comparisons between strains and conditions and facilitate integration with downstream statistical or machine learning analyses. By quantifying textural features of phenotypes, the proposed solution would move beyond binary classification towards a more comprehensive textural-characteristic profile. This shift from qualitative to quantitative analysis represents a critical step towards advancing both diagnostic and research applications in mycobacterial phenotyping.

1.3 Project objectives

1. Define requirements for automated quantitative analysis of mycobacterial cording phenotypes
2. Evaluate existing methods for image-based mycobacterial phenotyping, MODS assay analysis, and relevant image-processing techniques
3. Organize and standardize microscopy images of cording and non-cording, and disrupted for downstream processing
4. Develop image preprocessing techniques to account for acquisition artifacts and ensure uniformity across images.
5. Design an image-processing pipeline that automates detection, segmentation, and feature extraction of mycobacterial aggregates using appropriate algorithms.
6. Compute textural features to characterize cording phenotypes.

7. Use statistical analysis to evaluate differences in cording across strains, or experimental conditions.
8. Implement and test a machine learning model for automated categorization of phenotypes.
9. Evaluate system performance by assessing pipeline accuracy as compared to manual qualitative assessments.
10. Summarize findings, discuss limitations, and provide recommendations for future enhancements or applications.

1.4 Scope, limitations and assumptions

This project focuses on developing an automated image-processing pipeline for the quantitative analysis of mycobacterial cording phenotypes. The focus will be on dataset curation, pre-processing to correct acquisition artifacts, segmentation of bacterial aggregates, and extraction of textural features for comparison across strains. Additionally, a machine learning model will explore automated classification of phenotypes. The system aims to provide a reproducible and scalable method for quantifying cording phenotypes, moving beyond traditional qualitative assessments such as those used in MODS assays.

However, several limitations exist. The analysis depends on the quality and consistency of input images, and its generalizability may be constrained by the limited size and diversity of the available dataset. The pipeline will be optimized for the provided dataset and may require adaptation for other imaging modalities or laboratories. Finally, while machine learning will be explored, its performance will be constrained by the size and diversity of the available dataset.

1.5 Plan of development

This report is structured to explore the development of an image analysis pipeline for discriminating between cording and non-cording bacterial aggregates. The introduction establishes the background and significance of quantitative phenotype analysis in microbiology, highlighting the challenges of manual assessment and the potential of automated computational methods, while outlining the research objectives and scope. Building on this foundation, the literature review surveys existing approaches to image segmentation, texture feature extraction, and classification in biological imaging, identifying gaps in Mtb phenotyping. The methodology then details the integrated pipeline, encompassing automated segmentation for region isolation, extraction of texture descriptors, statistical evaluation of feature discriminability, and optimized machine learning classification using selected features for reproducible phenotype differentiation. Following this, the results section presents the empirical outcomes, including segmentation accuracy metrics, distributions of key texture features across phenotypes, and overall classification performance. The discussion interprets these findings in the context of biological implications, addresses limitations like dataset variability and computational efficiency, and compares results to prior studies to underscore contributions to automated aggregate analysis. The conclusion highlights the key achievements, and reaffirms the pipeline's efficiency in enabling objective, scalable phenotype discrimination and its broader applications in microbial research. Finally, the further work section proposes extensions that can potentially refine the pipeline.

Chapter 2

Literature Review

2.1 Biological and Diagnostic Context of Mycobacterial Cording

2.1.1 Background

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), remains one of the most significant infectious diseases globally. It primarily affects the lungs, spreading from person to person through airborne droplets. Despite being both preventable and curable, TB continues to pose a major public health challenge. According to the World Health Organization's Global Tuberculosis Report (2022), South Africa has an incidence rate of 468 cases per 100,000 population, a figure regarded as critically high. This burden is driven by several factors, including undernutrition, overcrowding, and co-morbidities such as HIV and diabetes, which complicate treatment and worsen disease progression (Mizrahi, 2024).

Early and accurate diagnosis is key to successfully treating TB. One of the biggest challenges faced in diagnosis of TB is the emergence of multi-drug resistant (MDR) (Rodriguez, et al., 2022) strains that are resistant to most first-line treatment. This poses a huge problem as administering the wrong drug can cause an increase in incidence of these MDR strains. Rapid and affordable TB diagnostic and drug susceptibility tests with high sensitivity are therefore necessary for early detection and personalized TB treatment. One such diagnostic tool is the Microscopic Observation Drug Susceptibility (MODS) assay, which enables simultaneous TB detection and drug susceptibility testing (Moore, et al., 2006).

2.1.2 Tuberculosis Diagnostic Methods

Accurate and timely diagnosis of Mtb infection is critical for reducing transmission and improving treatment outcomes. A range of diagnostic methods are available, each with advantages and limitations that influence their application in different settings.

Smear microscopy remains one of the oldest and most widely used techniques, offering low-cost and rapid results. However, it suffers from poor sensitivity, especially in paucibacillary and HIV co-infected patients (Steingart, et al., 2006). Culture-based methods, such as Löwenstein–Jensen and liquid media systems, are the gold standard for TB diagnosis, providing high sensitivity and allowing drug susceptibility testing. However, they are slow, with turnaround times ranging from days to several weeks. Molecular assays such as Xpert Mtb/RIF offer rapid and highly sensitive detection of Mtb and rifampicin resistance but are expensive, requiring advanced infrastructure (Boehme, et al., 2010).

The Microscopic-Observation Drug-Susceptibility (MODS) assay addresses several of these limitations. It combines culture and microscopy in a low-cost, liquid medium-based format, significantly reducing detection time compared to conventional culture. Importantly, MODS allows visual detection of characteristic cording phenotypes, which serve as a phenotypic readout of Mtb growth and can be used as a quick readout of drug resistance (Caviedes, et al., 2000). However, the assay currently relies on manual visual inspection of microscopy images, which is subjective and limits reproducibility.

2.1.3 Relevance of Cording to Tuberculosis Diagnosis

Cording refers to the formation of elongated, rope-like bundles of Mtb cells during growth in liquid culture (Middlebrook, et al., 1947). This phenotype is driven by the unique lipid-rich cell wall of Mtb called trehalose dimycolate (cord factor), and it shows the pathogen's ability to maintain structural integrity in a multicellular arrangement (Julián, et al., 2010). Cording is clinically significant as it has been associated with virulence, enhanced survival within host tissues, and altered drug susceptibility (Mishra, et al., 2023). Importantly, this structural phenotype provides a distinct visual biomarker that can be used for diagnostic and research applications.

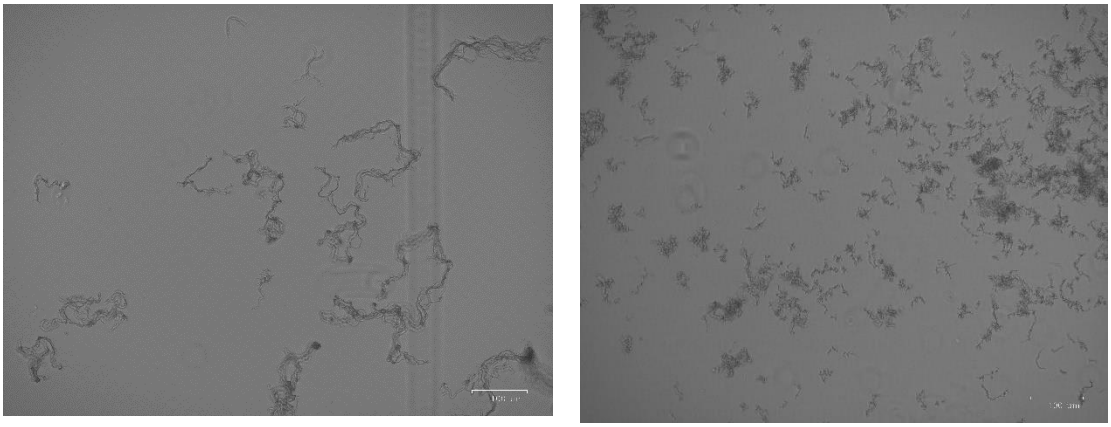
MODS assay leverages this property for early TB diagnosis. In MODS, patient samples are cultured in liquid media and inspected microscopically for the presence of cords, which signal Mtb growth (Figure 2.1(a)). This allows simultaneous detection of viable bacilli and assessment of drug resistance in a fraction of the time required for solid culture. However, the current reliance on manual visual inspection of cording is subjective, labour-intensive, and susceptible to inter-observer variability, limiting its scalability. Automating cording detection in MODS images offers a pathway to objective, reproducible, and high-throughput quantification, transforming the assay into a more robust diagnostic and research tool.

2.2 Traditional Approach to Cording Assessment

Traditionally, cording has been identified by light microscopy, using either bright-field or phase-contrast optics under low magnification. This method relies on morphological assessment, where trained observers distinguish cords based on their characteristic tightly wound, serpentine structures (McNerney, et al., 2012).

2.2.1 Cording Phenotype

Cording is characterized by the elongated, serpentine arrangement of bacterial cells, forming visually distinct parallel-aligned chains that coalesce into tightly wound, rope-like aggregates. This morphology arises from the interaction of mycolic acids in the cell wall, particularly through the virulence-associated lipid trehalose dimycolate (TDM, or cord factor). Cording is most pronounced in liquid cultures (e.g., MODS assays), where unrestricted growth preserves the structural integrity of these stranded aggregates (Caviedes, et al., 2000). In fluorescence microscopy, probes like DMN-Tre (a trehalose analog) selectively label mycolic acids (Swarts, et al., 2012)., revealing cords as organized, green-stained bundles due to their dense cell wall composition.



(a) Cording (b) Non-cording

Figure 2.1: Microphotograph used for the MODS assay captured using a Zoe Fluorescent Imager.

Figure 2.1(a) illustrates a microphotograph of Mtb in 7H9 liquid culture medium used for the MODS assay. The small colonies exhibit a characteristic strings-and-tangles morphology that is easily distinguishable from the non-cording colonies of *M. avium* and *M. kansasii* (Figure 2.1(b)). Non-cording phenotypes exhibit disorganized growth patterns, forming irregular clumps or amorphous aggregates without the characteristic parallel alignment of cording strains. Under microscopy, these clusters appear as granular or mesh-like masses, contrasting sharply with the serpentine structure of cording strains. This phenotype is frequently associated with cell wall modifications, such as alterations in mycolic acid composition or trehalose dimycolate (TDM/cord factor) production.

2.2.2 Limitations of Traditional Cording Assessment Approach

Manually assessing cording phenotypes presents several challenges. Firstly, subjectivity in interpretation can lead to inconsistent classification as some non-cording phonotypes may exhibit characteristic that might be mistaken for cording phenotypes. This means that accuracy would heavily rely on observer training and expertise. Secondly, low throughput restricts scalability and the generation of large datasets, as manual, microscopy-based screening is time-consuming and impractical for large-scale studies. Thirdly, quantitative limitations arise from the lack of standardized metrics for cord density, length, or degree of fragmentation, complicating comparative analyses. Additionally, strain variability such as subtle differences in cord tightness or density may be overlooked during visual inspection, particularly in strains with intermediate or disrupted cording.

2.3 Advances in Automated Mycobacterial Imaging

The presence of cords in liquid cultures has been used as a rapid and specific indicator of Mtb infection, particularly in the MODS assay (Caviedes, et al., 2000). Significant advances in imaging technologies and computational analysis have facilitated the automation of cording detection, reducing reliance on subjective interpretation, and improving diagnostic throughput.

2.3.1 Automated MODS plate readers

Early developments focused on automating the image acquisition process for MODS cultures. Comina et al. (2011) introduced an automated MODS plate reader that combined mechanical scanning with autofocus routines to capture high-resolution images of wells. This system enabled the consistent visualization of cording patterns while minimizing manual handling and biosafety risks. By standardizing image acquisition, this approach laid out the groundwork for subsequent computational analysis.

2.3.2 Classical Image Processing and Pattern Recognition

Classical computer vision methods formed the foundation for early automated cording detection. Alva et al. (2013) used morphological and textural features, combined with pattern-recognition algorithms such as Gray Level Matrices, to classify MODS images exhibiting the cording morphology. Features such as shape descriptors and intensity patterns were fed into classifiers including logistic regression and support vector machines. Reported classification accuracies were around 84–88%**,** depending on feature selection and dataset (Alva, et al., 2013).

In more recent work, (Hlabangana, 2024) employed a similar approach in which she used classical image approach to extract shape descriptors and textural features of cording phenotypes. In her work, a high dimensional feature array of shape descriptors (form factor, sphericity, roundness, solidity and elongation) and texture features (Haralick features specifically) were extracted and used to train a support vector machine to classify cording and non-cording phenotypes, achieving a 93% accuracy. Haralick features are statistical measures derived from a gray-level co-occurrence matrix (GLCM) that describe the texture of an image by quantifying patterns of pixel intensity relationships, such as contrast, homogeneity, and entropy. This research offered some insight into the potential usefulness of texture features in distinguishing different Mtb phenotypes.

2.3.3 Deep Learning Classification of Cording Patterns

(Lopez-Garnier, et al., 2019) introduced a convolutional neural network (CNN) framework for automated interpretation of MODS images, achieving high accuracy in detection of Mtb cords. Employing a VGG16-inspired 15-layer architecture trained on a diverse dataset of 12,510 grayscale images from three Peruvian laboratories, the model demonstrated robust performance with 96.63% accuracy, 94.74% sensitivity, and 97.83% specificity using 5-fold cross-validation, while cross-laboratory validation maintained ≥90% sensitivity and specificity despite variations in image quality. The CNN's learned features were visualized through gradient ascent and were observed to mirror expert diagnostic cues, such as sinuous cord structures and illumination contrasts, without requiring explicit segmentation.

2.3.4 Semantic Segmentation for Quantitative Analysis

(Núñez-Fernández, et al., 2020) advanced automated Mtb diagnostics by applying semantic segmentation to lens-free microscopy images within the framework of the MODS assay. The approach utilized a U-Net architecture to segment characteristic cord-like structures in grayscale images. The pipeline processed full images by dividing them into 150 smaller grayscale tiles (each paired with binary masks), performing segmentation on each tile, and reconstructing a comprehensive segmentation map. The model achieved a mean Intersection over Union (mIoU) of 0.86, a Dice coefficient of 0.91, and a pixel accuracy of 92.0%, demonstrating robust performance despite noisy inputs and ambiguous cord structures. These results highlight the feasibility of automated segmentation which enables quantitative analysis of cord morphology and texture which can support diagnostic interpretation and biological research.

2.3.5 Low-Cost Lens-Free Imaging with Machine Learning

(Solis, et al., 2016) evaluated a novel lens-free imaging system designed to enhance Mtb diagnostics through the MODS assay, offering a cost-effective alternative to traditional inverted microscopy. Utilizing a 24 mm² CMOS sensor, the system captured high-resolution images of MODS cultures at approximately 40× magnification in a single step, significantly reducing inspection time compared to manual methods (1–2 minutes per 24-well plate). Tested on ten sputum samples from the Universidad Peruana Cayetano Heredia in Lima, Peru, the lens-free imager produced images suitable for expert identification of Mtb cording patterns, demonstrating its potential for automating MODS interpretation in resource-limited settings. The system's ability to digitalize a wide field of view without expensive optical equipment addresses critical barriers in TB diagnostics, such as the need for skilled personnel and costly infrastructure, thereby supporting scalable TB control in high-burden regions.

2.3.6 Automated Time-Lapse Imaging of Cord Formation

(Kaslum, et al., 2021) introduced a high-throughput assay utilizing automated live-cell imaging to monitor phenotypic transitions in Mtb, focusing on distinguishing replicating versus non-replicating (persistent) bacterial forms. This approach employed time-lapse imaging coupled with software-driven quantification to track morphologies and growth dynamics in real time under varying conditions, such as antibiotic exposure or nutrient limitation. Validation experiments demonstrated the assay’s ability to discriminate phenotypic shifts. Time-lapse imaging reveals how aggregates form, grow, and respond to drugs, providing richer biological insights than static images.

2.3.7 Summary

Table 2.1: Pros and Limitations of Automated and Computational Approaches for MODS Image Analysis

Approach	Pros	Limitations
Automated MODS Plate Readers	Enables systematic and reproducible image acquisition. Reduces manual labour and biosafety risks. Provides images optimized for automated analysis.	High initial equipment cost and technical complexity. Image quality depends on optical system maintenance and calibration. No automated interpretation by itself.
Classical Image Processing and Pattern Recognition	Interpretable, transparent feature-based decision-making. Requires smaller training	Sensitive to lighting, background, and image variability. Limited generalizability across

Approach	Pros	Limitations
	datasets. Easier to implement in low-resource settings.	labs. Less robust for subtle or atypical cording patterns.
Deep Learning Classification (CNNs)	High accuracy and sensitivity. Robust to illumination and focus variation. Minimal manual feature engineering.	Performance depends on dataset size and diversity. Risk of overfitting to imaging conditions. Requires substantial computational resources.
Semantic Segmentation (U-Net, etc.)	Provides detailed spatial and quantitative morphology. Enables objective and reproducible cord measurements. Supports diagnostic and research applications.	Requires labour-intensive pixel-level annotations. Computationally demanding. Performance can degrade with poor contrast or artifacts.
Low-Cost Lens-Free Imaging with ML	Substantially reduces equipment costs. Captures wide fields of view for high-throughput screening. Suitable for decentralized and field diagnostics.	Lower resolution may obscure fine details. Requires ML models specifically trained for lens-free data. Needs periodic calibration in field settings.
Automated Time-Lapse Imaging	Captures dynamic cord formation and growth. Enables single-colony/single-cell analysis. Provides temporal data for virulence and drug response studies.	Requires biosafety containment and stable long-term imaging conditions. Produces large, storage-intensive datasets. Less suitable for routine diagnostics.

2.4 Image-Processing Techniques for Microscopic Images

Image processing plays a central role in converting raw microscopy data into quantitative biological insights. In the context of mycobacterial cording analysis, this involves detecting and segmenting bacterial aggregates, extracting textural and morphological features, and enabling phenotype classification. (Ghanegolmohammadi, et al., 2024) proposed a general pipeline that can be used in medical imaging which encompasses pre-processing, segmentation, feature extraction, and post-processing steps, each of which contributes to enhancing the quality and interpretability of microscopic data.

2.4.1 Pre-Processing Techniques

Pre-processing enhances image quality and prepares data for analysis by reducing noise, correcting uneven illumination, and improving contrast. Given the fine and often subtle structure of mycobacterial cords, tailored pre-processing steps must be carefully selected to preserve biologically meaningful features while improving visibility and contrast. Table 2.2 summarizes some of the pre-processing techniques can be applied (Gagniuc, 2025):

Table 2.2: Summary of Image Pre-processing Techniques for MODS Image Analysis

Pre-processing Step	Technique	Description / Application	Advantages	Limitations / Notes
Noise Reduction	Gaussian Filtering	Smooths random noise using a Gaussian kernel; useful for high-noise fluorescence images.	Simple and effective for global noise reduction.	May blur fine cord edges and small structures.
	Median Filtering	Replaces each pixel with the median of neighbouring pixels; ideal for brightfield images.	Preserves edges while removing salt-and-pepper noise.	Less effective for Gaussian or complex noise patterns.

Pre-processing Step	Technique	Description / Application	Advantages	Limitations / Notes
	Non-Local Means (NLM)	Denoises by averaging similar patches across the image; suitable for high-resolution or confocal images.	Preserves fine details and textures.	Computationally intensive; slower on large datasets.
Background Correction	Flat-Field Correction	Normalizes uneven illumination by dividing each image by a reference background.	Corrects illumination gradients in widefield microscopy.	Requires acquisition of a reliable background image.
	Morphological Subtraction (e.g., Rolling Ball Algorithm)	Removes large-scale background variations using morphological filtering.	Effective for phase-contrast images with bright backgrounds.	Parameter selection (e.g., ball size) affects accuracy.
Contrast Enhancement	Global Histogram Equalization	Redistributes pixel intensities to increase global contrast.	Simple and fast; enhances overall visibility.	Can distort intensity relationships; amplifies noise.
	CLAHE (Contrast-Limited Adaptive Histogram Equalization)	Enhances local contrast within small image tiles while limiting noise amplification.	Ideal for low-contrast brightfield images; preserves detail.	May introduce artificial contrast boundaries if over-applied.
	Gamma Correction	Applies nonlinear intensity scaling to enhance dim regions in fluorescence images.	Enhances visibility of faint cords without saturating bright areas.	Requires tuning of gamma value; excessive correction distorts brightness balance.

These pre-processing methods are essential for enhancing cording structures prior to segmentation and feature extraction. Carefully selecting and tuning these techniques ensures reliable and reproducible automated analysis, especially when dealing with variable imaging conditions or subtle phenotypic differences.

2.4.2 Segmentation Methods

Segmentation plays a pivotal role in the automated analysis of mycobacterial cording by isolating individual cords or aggregates from the image background and enabling feature extraction. However, several challenges complicate this process, including low contrast in brightfield images, overlapping or entangled cords, irregular morphologies (e.g., serpentine structures), and the presence of noise or artifacts such as debris and uneven staining.

Traditional segmentation methods can be applied to segment Mtb cording phenotypes (Rakesh, et al., 2018):

1. **Thresholding** techniques like Otsu's method and adaptive methods (e.g., Sauvola's) are fast and effective on high-contrast fluorescence images, but they often fail under uneven illumination or in low-contrast settings typical of brightfield microscopy.
2. **Edge-based methods**, such as Canny edge detection or Sobel filters, may be useful for preserving thin boundaries of cords, but may struggle to form closed contours in noisy environments.
3. **Region-based approaches**, particularly the watershed algorithm, could be effective at separating clumped cords using intensity gradients, although they are prone to over-segmentation.
4. **Morphological operations** like erosion, dilation, and opening/closing can be employed post-thresholding to clean binary masks but must be used cautiously to avoid distorting delicate cord structures.

To address these limitations, machine learning (ML)-based segmentation methods can now be employed. Núñez-Fernández et al., (2020) successfully implemented a **deep learning** method, particularly a U-Net convolutional neural network architecture, to segment Mtb cords and achieved a high Dice Coefficient of 0.91. This good performance is

due to the ability of neural networks to learn context-aware features and deal with noise, intensity variation, and overlapping cords.

These segmentation strategies form the foundation for quantifying structural features of mycobacterial cords and are critical for enabling reproducible phenotype classification in automated diagnostic pipelines.

2.4.3 Texture Feature Analysis of Mtb Phenotypes

Following image segmentation, feature extraction is employed to quantitatively characterize mycobacterial cords and aggregates. In the context of Mtb research, texture analysis has been employed to detect morphological phenotypes such as cording in MODS cultures, microcolony patterns, and growth heterogeneity. Classical texture descriptors remain valuable because they are interpretable, computationally efficient, and effective in low-data settings. They are often used either as primary features for classification or as complementary inputs to machine learning models (Alva et al., 2013; Hlabangana 2024).

Several categories of texture feature extraction methods can be applied to Mtb images. **Statistical methods**: such as Gray Level Co-occurrence Matrices (GLCM), Gray Level Run Length Matrices (GLRLM), and Local Binary Patterns (LBP), can quantify intensity relationships between neighboring pixels to capture spatial texture. **Transform-based methods,** including Gabor filters and wavelet transforms, represent texture information in the frequency or multi-resolution domain, which can be useful for capturing repetitive cord-like structures at different scales (Armi & Fekri-Ershad, 2019). Some studies have combined multiple methods to improve classification performance (Alva, et al., 2013).

Table 2.3: Comparative analysis of Texture Feature extraction that can be applied to Mtb microscope images

Method	Principle	Applied to MODS?	Pros	Limitations	Reported Performance
Gray Level Co-occurrence Matrix (GLCM)	Statistical measure of how often pixel pairs with specific intensity values occur at defined spatial relationships; yields Haralick features	Yes	Interpretable; captures local spatial texture; works well on non-corded patterns	Sensitive to noise and illumination.	Hlabangana. (2024): ~93% classification accuracy for cord vs non-cord when combined with shape descriptors
Gray Level Run Length Matrix (GLRLM)	Measures length of consecutive pixels with same intensity in specific directions	No published MODS studies	Can be effective for detecting elongated cord-like structures	Requires discretisation of intensity values; may be affected by uneven staining	N/A
Local Binary Patterns (LBP)	Encodes local texture by thresholding neighbourhood pixels relative to centre pixel	No published MODS studies	Rotation invariant variants available; fast computation; robust to monotonic intensity changes	Sensitive to noise in homogeneous regions; small neighbourhoods may miss large-scale structures	Not reported for Mtb, but can be combined with GLCM for improved texture discrimination
Gabor Filters	Band-pass filters in multiple scales and orientations to capture frequency and directional texture information	No published MODS studies	Excellent at detecting more uniform textures (e.g., non-cords); robust to	Requires careful selection of scales/orientations; higher computational cost	N/A

Method	Principle	Applied to MODS?	Pros	Limitations	Reported Performance
			illumination changes		
Wavelet Transform	Decomposes image into multi-resolution sub-bands to capture textures at different scales	Yes	Multi-scale analysis well suited for variable cord thickness; compact representation	Choice of wavelet basis impacts performance; edge effects can occur	Alva et al. (2013): wavelet + GLCM combination achieved up to 90% accuracy for MODS image classification

Among these methods, GLCM features are the most widely used for Mtb cording detection, particularly in MODS image analysis (Alva et al., 2013, Hlabangana 2024). They offer a good balance of interpretability and performance, especially when combined with simple classifiers such as logistic regression or support vector machines. GLRLM features could potentially be suitable for characterizing the elongated, oriented nature of cording phenotypes, while Gabor filters and wavelet transforms capture multi-scale and could potentially be suitable for capturing non-cording phenotypes. Although LBP is computationally efficient, it has been less frequently applied in isolation to Mtb phenotyping.

Texture features can also be integrated into modern deep learning workflows, for instance, as handcrafted features alongside CNN features to improve interpretability, or for model regularization in low-data settings. While deep learning methods now dominate biomedical image analysis, classical texture descriptors remain relevant in Mtb phenotyping for explainable, resource-efficient, and interpretable diagnostic tools, especially in low-resource laboratories.

2.4.4 Quantitative Analysis of Texture Features

Statistical analysis is essential in image-based phenotyping because it enables the quantification and validation of morphological or textural differences across experimental conditions. Statistical tests such as analysis of variance (ANOVA) and non-parametric alternatives allow researchers to assess whether extracted features (for example, texture descriptors) differ significantly between phenotypic groups, while applying Student t-test for comparisons (Julián, et al., 2010). The foundational work of (Haralick, 1973) demonstrated that texture features derived from the gray-level co-occurrence matrix (GLCM) can classify images with ~82-89% accuracy across different image types. More recently, (Zayed, 2015) applied Haralick texture features in biomedical imaging and used statistical tests to identify texture features with significant discriminative power between images from different conditions. After statistical filtering of relevant features, machine-learning classifiers (e.g., Random Forests) can further automate phenotype categorization, improving reproducibility and scalability of phenotyping workflows.

2.5 Rationale for Developing an Automated Cording Analysis Pipeline

Although significant progress has been made in automating different components of MODS image analysis: ranging from plate reading to deep learning classification and texture-based feature extraction, important gaps remain that hinder the development of fully integrated, scalable diagnostic tools. First, most existing studies focus on isolated components of the pipeline (e.g., acquisition or, segmentation) rather than on end-to-end frameworks that combine pre-processing, segmentation, feature extraction, statistical analysis, and classification (Alva et al., 2013; Comina et al., 2011; Núñez-Fernández et al., 2021). This fragmentation limits reproducibility and complicates deployment in real diagnostic settings. Second, while deep learning has shown high accuracy for cording classification (Lopez-Garnier, et al., 2019) these methods often function as “black boxes” and lack transparency making it difficult to link outputs to biologically meaningful texture or morphological features. Classical texture analysis methods offer

interpretability but are underutilised in recent MODS-focused research, with techniques such as GLRLM, LBP, and Gabor filters remaining largely unexplored for Mtb phenotyping despite their success in other biomedical imaging domains when combined with GLCM features (Dheepak, 2024). Third, there is a lack of standardised, quantitative measures for cording phenotypes that would allow objective comparisons across laboratories, as manual visual inspection remains the standard (McNerney, et al., 2012). Furthermore, post-processing statistical analysis, though shown to be powerful in identifying discriminating features in medical images (Zayed, 2015), has not been systematically integrated with automated pipelines to support phenotype classification or hypothesis testing.

Given these gaps, there is a need to develop a fully automated, statistically informed image analysis pipeline for MODS cultures that combines pre-processing, accurate segmentation, texture-based quantitative analysis, and explainable classification. Such a pipeline would reduce inter-observer variability, improve reproducibility, and increase throughput, enabling standardised, objective, and scalable TB diagnosis. Moreover, integrating statistical analysis of texture features into the pipeline would lay the foundation to establishing quantifiable metrics that can be used to characterize Mtb phenotypes.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the details of the methodological framework used to develop a fully automated image analysis pipeline for the detection and quantitative assessment of mycobacterial cording phenotypes in cultures that discriminate cording from non-cording variants. The methodology involves a structured, sequential pipeline with five critical stages: data pre-processing, weakly-supervised image segmentation, multi-scale texture feature extraction, statistical analysis of phenotypic signatures, and machine learning classification. Each stage was designed to address specific challenges in mycobacterial image analysis, including uneven illumination artifacts, limited dataset, the absence of pixel-perfect ground truth annotated masks, high-dimensional feature spaces, and the need for statistical validation.

The pipeline (Figure 3.1) begins with standardized images captured from a fluorescence microscopy and then image enhancement techniques were applied to mitigate artifacts arising from acquisition, such as uneven illumination, which improved image quality. The segmentation phase employed an innovative Edge-Dilated Weak Masking (EDWM) strategy, which eliminated the need for labour-intensive manual annotations while preserving biological details in delineating phenotype structures. For feature extraction, a dual approach was utilized, combining Gray Level Co-occurrence Matrix (GLCM) analysis with Local Binary Pattern (LBP) methodologies to capture phenotypic characteristics across multiple spatial scales. Finally, the classification stage employed a Support Vector Machines (SVM) to transform these quantitative features into reliable phenotype predictions.

This chapter is structured to ensure reproducibility of the research methodology. Section 3.2 details the data collection and curation process, which involves specifications for image acquisition and data folder structure. Section 3.3 describes the pre-processing techniques that were implemented to enhance image quality while preserving critical biological structures. Section 3.4 presents the segmentation methodology, with a focus on the U-Net architecture and the weak supervision strategy that was employed. Section 3.5 explains the feature extraction process that was utilized to capture phenotypic characteristics. Section 3.6 outlines the statistical analysis and classification approaches, including the use of a Support Vector Machine for phenotype prediction. While some tools and packages are loosely mentioned throughout the methodology, Section 3.7 is a dedicated section that specifies the computational environment and all the tools used to implement the pipeline. Finally, Section 3.7 outlines the experimental design and validation strategies.

The methodological framework proposed in this chapter supports the findings reported in Chapter 4 and provides a foundation for future research in automated Mtb phenotype analysis. Each technical decision, parameter selection, and implementation details were documented in this section to ensure that the pipeline can be accurately reproduced, validated, and extended by other researchers in the field.

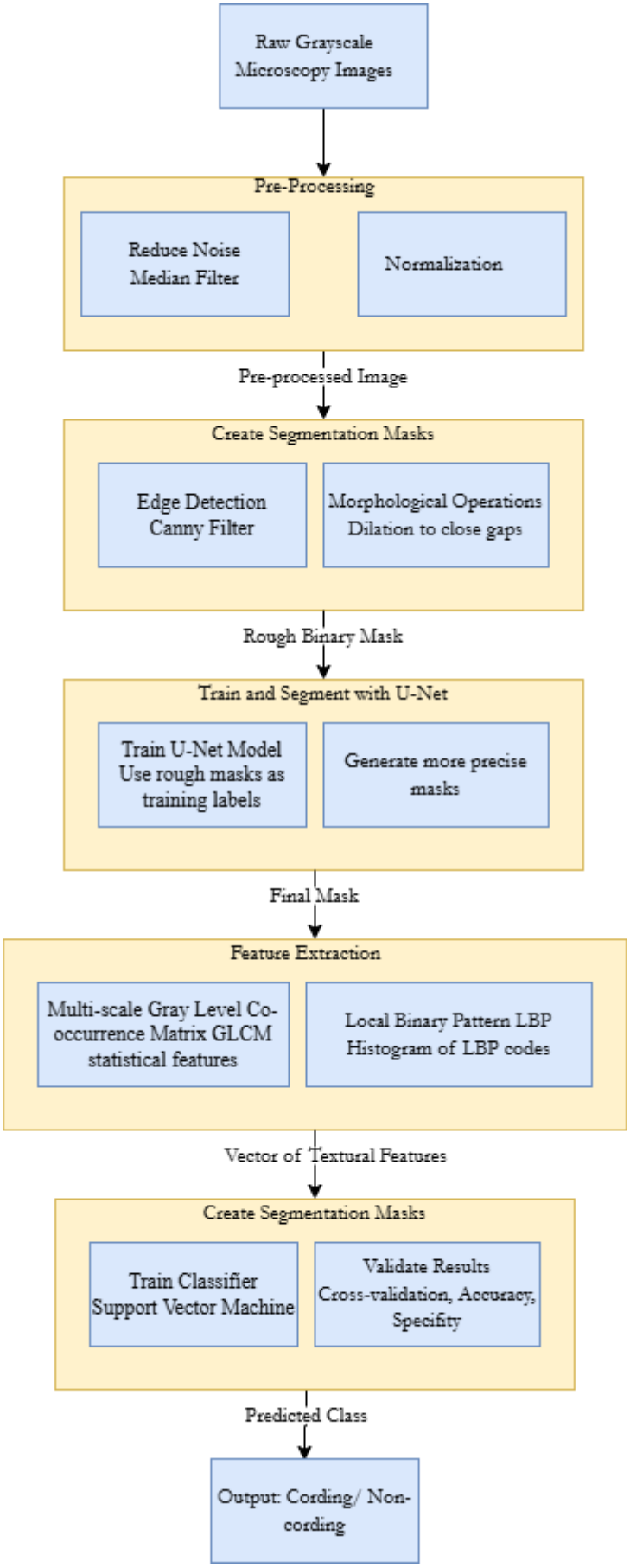


Figure 3.1: Proposed Pipeline for automated quantitative analysis of Mtb cording phenotypes

3.2 Dataset Curation

3.2.1 Dataset Description

The dataset for this study consisted of 183 images of Mtb cultures, acquired using a ZOE Fluorescent Cell Imager at 20× magnification with a resolution of 2592×1944 pixels. These images were obtained from a controlled laboratory environment focused on studying mechanopathology of Mtb in ongoing research by Dr MK Mason (personal communication, 2025). The dataset was strategically organized into a structured directory with standardized filenames to facilitate processing. It included 15 folders, each containing 9 images each from various wells, (totaling 135 images) exhibiting variations of either cording or non-con-cording phenotypes. These images were not annotated therefore; they were regarded as unlabeled data. Two separate folders, Parental_Phigh (a cording variant) and Parental_Pminus (a non-cording variant) were also provided (each consisting of 24 images) with control strains for cording and non-cording phenotypes respectively. The control folders were used as labeled data.

3.2.2 Dataset Rationale

The folder structure was designed to balance variability with controlled phenotype representation. The 15 folders, each with 9 images, captured diverse variations of cording and non-cording phenotypes, providing a broad sample for testing the pipeline's generalizability. The two control folders, with 24 images each for Parental_Phigh (cording) and Parental_Pminus (non-cording), were established as a baseline dataset for training and validation, leveraging the known phenotypic labels to guide model development.

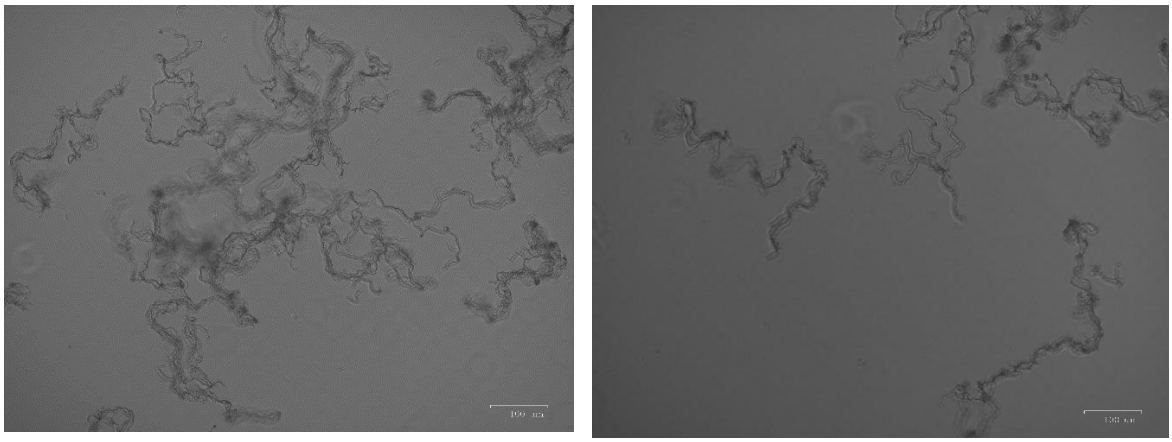
3.2.3 Dataset Inspection

An initial visual assessment of the dataset provided the following insights:

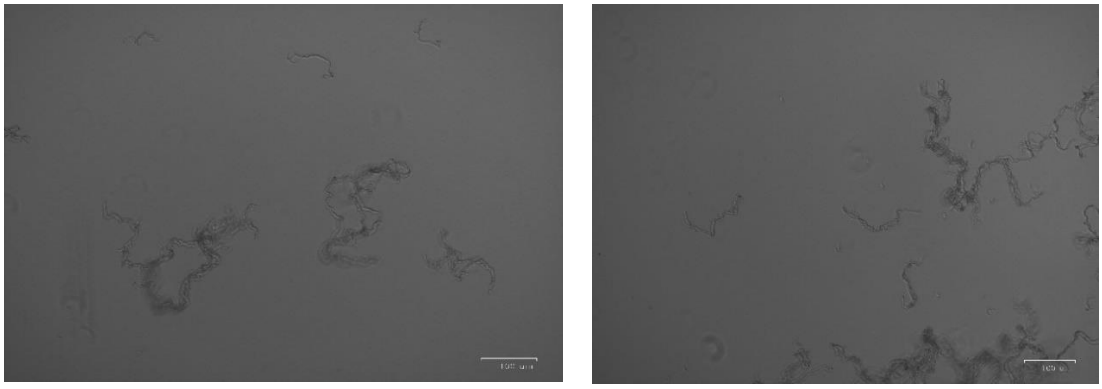
Cording Images: These images displayed thin, elongated, and branching fibrous structures. From Figure 3.2, with noticeable noise, possibly from the microscope reflections. Uneven illumination is evident in some of the images, suggesting variability in lighting during acquisition. This unevenness may affect boundary definition, though the overall structure of the cords remains distinguishable.

Non-Cording Phenotype Images: These images show less organized aggregates, with a granular texture that contrasts with the filamentous nature of the cording phenotype (Figure 3.3). Noise levels appear slightly higher, potentially complicating segmentation. Uneven illumination is also present in some of the images, similar to the cording images.

These observations suggested that pre-processing steps, specifically noise reduction and illumination normalization were essential to enhance quality of both phenotype images for accurate segmentation and feature extraction.

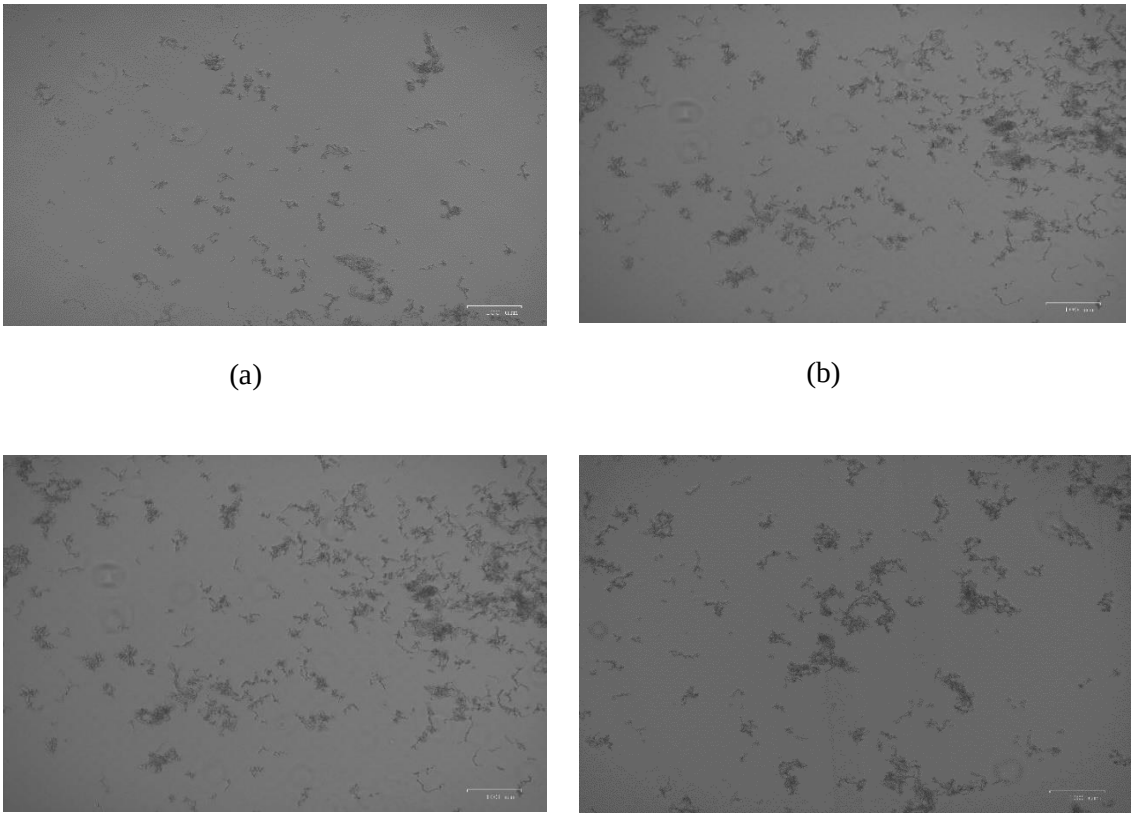


(a) (b)



(c) (d)

Figure 3.2: Images (a), (b), (c) and (d) showing cording strain



(a) (b)

(c) (d)

Figure 3.3: Images (a), (b), (c) and (d) showing non-cording strain

3.3 Image Pre-Processing

3.3.1 Noise Reduction

Noise reduction was the initial and critical step to improve the quality of the 183 microscopy images prior to feature extraction. A 3×3 median filter was implemented using the `OpenCV` library (v4.5, Python) to smooth random noise while preserving the edges of bacterial aggregates. Each pixel intensity was replaced by the median of its 3×3 neighborhood, effectively suppressing high-frequency artifacts without distorting cord boundaries.

Across the 17 image folders, the Root Mean Square (RMS) contrast, a measure of intensity variation, decreased by an average of 12.4% (range 0.75–1.09), confirming effective noise suppression while maintaining phenotypic integrity. RMS contrast was chosen as a quantitative metric due to its sensitivity to local intensity fluctuations, and was calculated as follows:

$$RMS\ Contrast = \sqrt{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I_{i,j} - \bar{I})^2}, \quad (3.1)$$

where Iij represents the intensity of the pixel at position (i, j), I is the mean intensity across the image, and M × N is the image resolution (2592x1944 pixels).

Raw RMS contrast values ranged from 6.67 (Ci-E06-2) to 8.49 (Ci-E06-8) and were reduced to filtered values between 5.80 (Ci-NT-4) and 7.44 (Ci-U17-4), as detailed in Table 4.1 (Chapter 4). The selection of a 3×3 kernel represented an optimal trade-off between noise suppression and edge preservation, verified through both visual inspection and contrast-based evaluation.

3.3.2 Artifact Correction

Uneven illumination and intensity gradients were corrected to ensure uniform image brightness across the dataset. A global z-score normalization approach was implemented using `NumPy`, where each image's pixel intensities were normalized based on the global mean and standard deviation, then rescaled to the original 0–255 intensity range. This process standardized contrast across images and mitigated artefactual lighting variations, enabling consistent downstream texture analysis.

3.3.3 Validation

Pre-processing performance was validated using a combination of quantitative and qualitative assessments. The results are presented in Chapter 4.2. Quantitatively, the RMS contrast values provided a numerical measure of noise suppression, with reductions per folder ranging between 0.75 (Ci-U17-3) and 1.09 (Ci-E06-8). Qualitatively, side-by-side comparisons of raw and filtered images confirmed that the aggregate structures were preserved after filtering. Visual inspection ensured that the median filter effectively removed random noise without degrading morphological integrity, an essential requirement for subsequent segmentation and feature extraction stages.

3.3.4 Design Considerations

The median filter was selected over alternatives such as Gaussian blur and Non-Local Means (NLM) filtering due to its ability to preserve sharp edges while effectively removing impulse (salt-and-pepper) noise common in brightfield microscopy. Although Gaussian filters are effective for general denoising, they can over-smooth fine cord structures, while NLM filtering, though highly accurate, is computationally more intensive and less practical for high-throughput preprocessing (Gagniuc, 2025).

RMS contrast was used as the primary quantitative noise metric because it directly measures overall intensity variation, making it more sensitive to distributed noise patterns than Peak Signal-to-Noise Ratio (PSNR), which primarily captures peak deviations. These methodological choices ensure a balance between computational efficiency, noise reduction, and preservation of phenotypic detail necessary for reliable downstream texture analysis.

3.3.5 Summary

Table 3.1: Summary of Pre-Processing Rationale

Step	Method	Purpose	Rationale
Noise Reduction	3×3 Median Filter (`OpenCV v4.5`)	Suppress random noise while preserving cords	Balances denoising and edge retention
Artifact Correction	Z-Score Normalization (`NumPy`)	Correct illumination variation	Standardizes intensity distribution
Validation	RMS Contrast & Visual Inspection	Quantify noise suppression and structure preservation	Ensures integrity of phenotypic detail

3.4 Image Segmentation

Image segmentation was the next critical step in the pipeline. The goal for segmentation was to isolate bacterial aggregates for subsequent texture analysis and classification. A deep learning U-Net architecture was proposed to perform the segmentation task which would leverage a supervised learning approach to isolate bacterial aggregates. Due to the absence of pre-annotated segmentation masks, a weak supervision strategy was adopted to generate coarse labels that would allow the training of a U-Net segmentation model.

3.4.1 Data Description

The U-Net was trained using data from the control strains of cording (Parental_Phigh) and non-cording (Parental_Pminus) phenotypes, each with 24 images totaling 48 images. These images were selected because they represented the phenotypic characteristics, providing a reliable baseline for model training. This ensured a balanced representation of cording and non-cording phenotypes, which was critical for the model to learn discriminative features despite the limited sample size. Generalization of the model was assessed on a held-out dataset of 135 images.

3.4.2 Weak Label Generation

U-Net model requires ground truth annotation masks to perform segmentation, which were not available for the given dataset. Traditional global thresholding methods such as Otsu's method were initially applied to create binary masks. However, these approaches failed to consistently capture relevant structures due to uneven illumination and low contrast at image edges, leading to inaccurate segmentation boundaries as reported in Section 4.3.1 of Chapter 4.

To overcome these challenges, a novel weakly supervised mask generation approach termed Edge Dilated Weak Masking (EDWM) was developed. This method leverages edge detection followed by morphological dilation to generate coarse but spatially meaningful masks that preserve morphological edge structure. The EDWM approach automatically generates approximate but spatially meaningful masks by enhancing edge-based structural cues and expanding them into segmentation regions. This method was specifically designed to capture bacterial aggregate structure boundaries while minimizing manual annotation requirements.

The EDWM technique was implemented in Python using the `OpenCV` (v4.5), `NumPy`, and `SciPy` libraries, and consists of the following sequential stages:

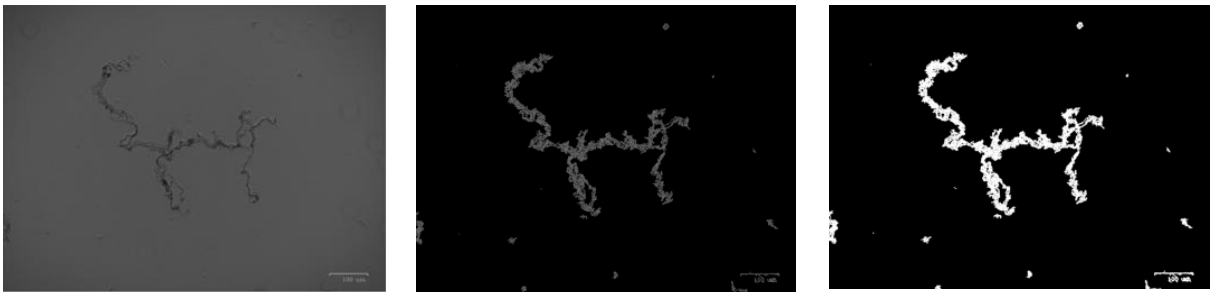
1. **Grayscale Conversion and Normalization**
   This ensured that all images were converted to grayscale using `OpenCV`'s cv2.cvtColor() function. It also ensured that all pixel values were normalized to an 8-bit range (0–255) using z-score normalization. This step preserves relative contrast while standardizing dynamic range for consistent edge detection.

2. **Edge Detection**
   Structural boundaries corresponding to bacterial aggregates and cords were extracted using the **Canny edge detector** (`cv2.Canny()`), with empirically optimized threshold values of 10 (lower) and 100

(upper). The Canny operator was selected for its ability to detect fine, high-gradient transitions characteristic of cord boundaries while suppressing noise-induced edges.

3. **Morphological Dilation**
   To convert thin edges into spatially extended regions suitable for weak mask generation, a morphological dilation operation was applied using an elliptical structuring element `(cv2.getStructuringElement(cv2.MORPH_ELLIPSE))` of kernel size 3×3. Dilation was iterated twice (`iterations = 2`), effectively expanding edge boundaries into continuous regions approximating bacterial clusters. This process preserved the geometric outline of aggregates while filling small discontinuities.

4. **Output Mask Generation**
   The resulting dilated binary masks were saved as images in a separate output directory, with filenames automatically appended with the suffix *mask*. The process was applied iteratively to all images within the input dataset, producing a complete set of weak labels corresponding to the original images.



(a) Original Image (b) Canny edge (c) Dilated

Figure 3.4: The EDWM method successfully generated binary coarse masks with well-maintained edge integrity

3.4.3 U-Net Architecture

The segmentation model was based on the U-Net architecture. The model featured an encoder-decoder structure with four downsampling and upsampling stages, connected by skip connections to preserve spatial details. The model's architecture is summarized below:

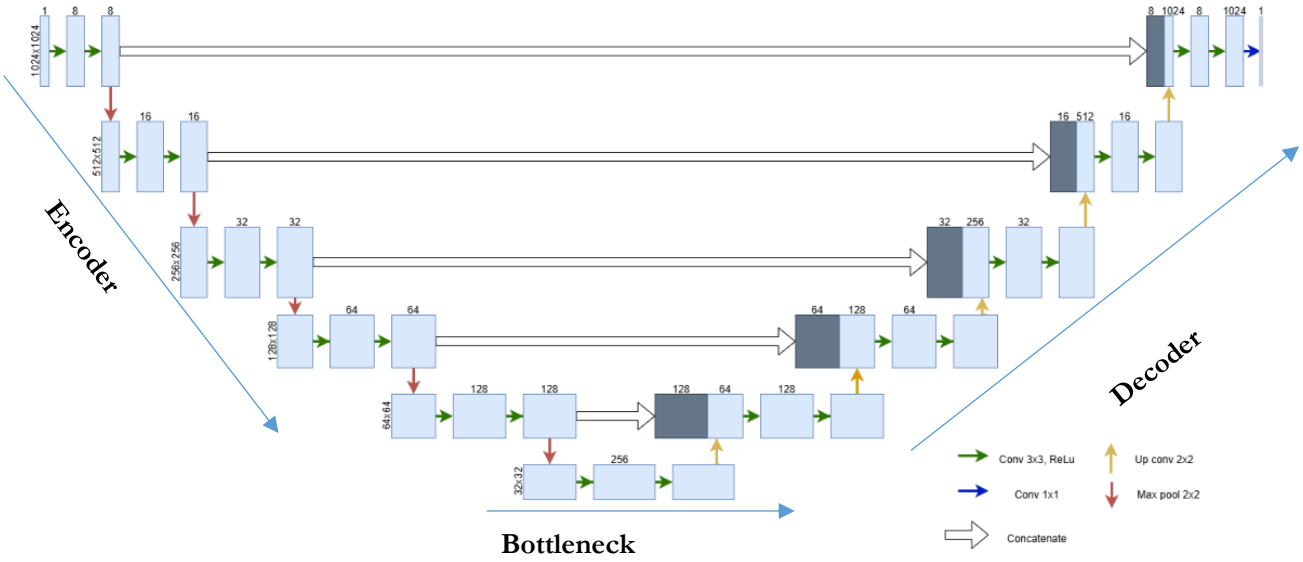


Figure 3.5: U-Net Architecture employed for segmentation

1. **Encoder**
   The encoder path comprised four levels of convolutional blocks (two 3×3 convolutions with ReLU activations) followed by 2×2 max pooling, with input dimensions of 1024×1024×1 (grayscale). The number of feature channels increased progressively (16 → 32 → 64 → 128) to enable the network to capture increasingly abstract spatial features.

2. **Bottleneck**
   At the network's deepest layer, a double 3×3 convolution with 256 filters served as a bottleneck, enabling rich feature abstraction at the lowest resolution.

3. **Decoder**
   The decoder path mirrored the encoder, employing bilinear upsampling, concatenation with corresponding encoder feature maps (skip connections), and double 3×3 convolutions to reconstruct spatial details.

4. The final layer was a 1×1 convolution producing three output channels (logits), corresponding to the background, cording, and non-cording classes.

3.4.4 Training Protocol and Optimization

The training protocol was designed to address class imbalance and ensure generalization, with the following components:

A. ***DATA PREPARATION***
A single image was 2592×1944 ≈ 5 million pixels, which would require an enormous GPU memory to train a full resolution image. Due to memory constraints, each original image and its corresponding EDWM-generated mask were tiled into 1024×1024 patches with a 25% overlap. The overlap ensured coverage of morphological structures that crossed tile boundaries.

Three classes were defined:
- **Class 0:** Background (no bacterial structure)
- **Class 1:** Cording (Parental_Phigh)
- **Class 2:** Non-cording (Parental_Pminus)

This process generated 288 tiles, with 143 for cording and 144 for non-cording. The dataset was partitioned into 80% (230 tiles) training and 20% (58 tiles) validation subsets using a stratified approach. There were no overlaps between training and validation sets.

B. LOSS FUNCTION AND OPTIMIZATION
A weighted categorical cross-entropy loss was used with class weights of 0.67, 1.34, and 1.33 for background, cording, and non-cording, respectively, to mitigate class imbalance. The Adam optimizer was employed with an initial learning rate of 1×10^{-4}, $\beta1 = 0.9$, and $\beta2 = 0.999$. Size filtering was implemented to remove small artifacts (areas <100 pixels). This preserved biologically relevant aggregates while eliminating noise-induced fragments. A learning rate scheduler reduced the learning rate by a factor of 0.5 upon validation loss plateau to optimize convergence.

C. TRAINING CONFIGUARATION
Training was conducted for 100 epochs, with early stopping after 15 epochs of no improvement in validation loss. Validation was performed for every epoch, thereby computing comprehensive performance metrics. A batch size of 2 was used due to constrained by GPU memory limitations.

3.4.5 Performance Assessment Metrics

The quality of the segmentation methodology was evaluated using a comprehensive set of metrics to ensure reliable assessment of the U-Net model's performance in identifying aggregate regions. These metrics were selected to balance the practical constraints of limited annotations with the analytical need for precise delineation of cording and non-cording aggregates.

Two primary metrics were employed to quantify segmentation performance:

1. Dice Coefficient (DC): The DC measured the overlap between the predicted segmentation mask (X) and the ground truth or weak label mask (Y). It was calculated as:

$$DC = \frac{2|X \cup Y|}{|X|+|Y|} \quad (3.2)$$

where |X∩Y | represents the number of pixels in the intersection of the predicted and ground truth masks, and |X| and |Y | denote the number of pixels in the predicted and ground truth masks, respectively. The DC ranges from 0 to 1, with higher values indicating better segmentation accuracy.

2. Intersection over Union (IoU): The IoU, also known as the Jaccard index, quantified the ratio of the intersection to the union of the predicted and ground truth masks. It was computed as:

$$IoU = \frac{|X \cap Y|}{|X \cup Y} \quad (3.3)$$

where |X ∪ Y | represents the number of pixels in the union of the two masks. Like the DC, IoU ranges from 0 to 1, with higher values reflecting greater overlap and segmentation precision.

To account for phenotype diversity, class-specific evaluations were conducted:

- Per-Class Dice and IoU: Dice and IoU metrics were calculated separately for cording and non-cording phenotypes to assess segmentation performance across distinct classes. This ensured that the model effectively captured the unique morphological characteristics of each phenotype.
- Probability Calibration Assessment: Probability maps were generated to assess the calibration of probability outputs from the U-Net model, ensuring that predicted probabilities aligned with actual segmentation outcomes.

Qualitative evaluations from the reconstructed masks complemented the quantitative metrics to verify the segmentation results:

- Structural Continuity Evaluation: The continuity of cording phenotypes was assessed to ensure that segmented regions preserved the characteristic elongated and clustered morphologies.
- Boundary Smoothness and Artifact Analysis: Segmentation outputs were examined for boundary smoothness and the presence of artifacts

3.4.6 Design Strategy

The U-Net architecture was selected for its proven success in Mtb cord segmentation by (Núñez-Fernández, et al., 2020). The symmetric encoder-decoder structure with skip connections enables precise localization of irregular aggregate boundaries while maintaining contextual understanding of spatial relationships. This design is particularly suited for capturing the heterogeneous morphology and size variability of the two phenotypes, even with limited annotated data. EDWM was proposed as a novel, cost-effective weak supervision method to avoid the need for extensive manual annotations. The method leveraged edge information to generate coarse masks which saved annotation time, given the limited duration to complete this project. A 1024x1024 tile size was chosen to balance computational feasibility and maintenance of segmentation detail. A smaller tile size might have resulted in some in the model struggling to differentiate between cording and non-cording phenotypes which could lead to biased, inaccurate segmentation. The model was used to segment the 135 unseen images for downstream analysis. This strategy supported the pipeline's scalability and aligned with the project's goal of automating phenotype analysis across diverse, limited datasets.

3.4.7 Summary

The segmentation methodology effectively addressed the challenge of limited annotations while achieving precise aggregate region identification (Section 4.3.2). The EDWM strategy, combined with the U-Net architecture and rigorous training protocol, provided a foundation for subsequent texture feature extraction and phenotypic characterization, which ensured reliable performance for downstream analyses.

3.5 Feature Extraction

Following the segmentation phase, a comprehensive multi-scale texture analysis was conducted to quantitatively characterize structural differences between cording and non-cording phenotypes. A custom Python pipeline was developed, integrating Gray Level Co-occurrence Matrix (GLCM) features, Local Binary Patterns (LBP), and regional intensity statistics to generate a comprehensive feature set. This section provides a detailed description of the feature extraction methodology.

3.5.1 Multi-Scale GLCM Features

GLCM features were extracted at varying spatial scales to capture textural patterns of cording and non-cording phenotypes. Gray-Level Co-Occurrence Matrix (GLCM) is a second-order statistical method that captures spatial relationships between pixel intensities within a given image. It is particularly well-suited for identifying patterns related to structural arrangement, granularity, and heterogeneity in grayscale images.

A. DEFINITION

The GLCM, denoted as $P(i, j|d, \theta)$, represents the probability of a pixel with gray level i being adjacent to a pixel with gray level j, given a distance d and angle θ. For an image with G gray levels, the GLCM is a $G \times G$ matrix (Armi & Fekri-Ershad, 2019).

$$P(i,j|d,\theta) = \sum_{x=1}^{M}\sum_{y=1}^{N}\begin{cases}1, & if\ I(x,y=i\ and\ i(x+\Delta x,y+\Delta y)=j \\ 0, & otherwise\end{cases}, \tag{3.4}$$

where (x,y) denotes the coordinates of a reference pixel, (x+Δx, y+Δy) is the neighbor pixel at distance d and angle θ, Δx=dcosθ and Δy=dsinθ, i,j ϵ {0, 1,…,G-1}, and M and N are the image dimensions.

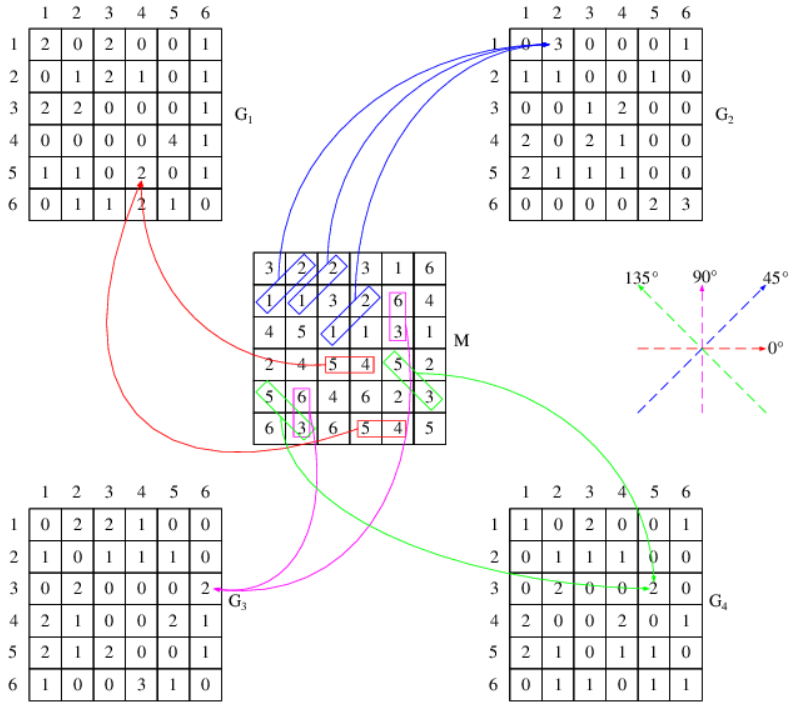


Figure 3.6: GLCM constructs a matrix that records how often pairs of pixel values (gray levels) occur at a specified distance and orientation. Adapted from (Vyas, 2017)

B. PARAMETER SELECTION

All grayscale images were first normalized to the 0–255 range and then quantized to 64 gray levels to reduce computational complexity while preserving meaningful intensity variations. GLCMs were computed for multiple distances to capture texture information at different spatial scales:

- **Medium scale**: d=3, d=4 and d=5 pixels
- **Macro scale**: d=7 pixels

For each distance, GLCMs were computed at two orientations, θ=0° (horizontal) and θ=45° (diagonal). The decision to use 0° and 45° angles was based on a combination of computational efficiency and biological relevance. The 0° angle captures horizontal texture patterns, while the 45° angle accounts for diagonal variations, which may highlight subtle organizational differences in both cording and non-cording aggregates. This selection reduces computational overhead for the dataset, optimizing processing on limited hardware, while avoiding redundancy from symmetric orientations. This multi-orientation analysis helps capture phenotype structures that may appear at different angles due to cord orientation.

C. HARALICK FEATURE COMPUTATION

From GLCM, 6 Haralick texture features, which summarize different aspects of intensity co-occurrence distribution were calculated:

1. Contrast: Measures intensity variation features between neighboring pixels

$$Contrast = \sum_{i=0}^{G-1}\sum_{j=0}^{G-1}|i-j|^2 P(i,j) \quad (3.5)$$

2. Dissimilarity: A linear measure of intensity differences

$$Dissimilarity = \sum_{i=0}^{G-1}\sum_{j=0}^{G-1}|i-j| P(i,j) \quad (3.6)$$

3. Homogeneity: Measures closeness of the distribution to the diagonal (high for smooth textures)

$$Homogenity = \sum_{i=0}^{G-1}\sum_{j=0}^{G-1}\frac{P(i,j)}{1+(i-j)^2} \quad (3.7)$$

4. Energy: Measures textural uniformity (angular second moment (ASM))

$$Energy = \sum_{i=i=0}^{G-1}\sum_{j=0}^{G-1}[P(i,j)]^2 \quad (3.8)$$

5. Correlation: Measures textural uniformity (angular second moment (ASM))

$$Correlation = \frac{\sum_{i,j}(i-\mu_i)(j-\mu_j)}{\sigma_i\sigma_j}, \quad (3.9)$$

where μ_j, μ_j and σ_i, σ_j are the means and standard deviations of the marginal probabilities.

6. Angular Second Moment: Reported separately from Energy, but mathematically equivalent (Equation 3.8).

These features were computed for each distance and orientation. The resulting number of GLCM features was therefore:

Table 3.2: Summary of total GLCM features extracted

Scale	Distances (Pixels)	Orientations	No. of Properties	Total Features
Medium-scale	3, 4, 5	0^0, 45^0	6 (contrast, correlation, energy, homogeneity, entropy, variance)	36
Macro-scale	7	0^0, 45^0	6 (contrast, correlation, energy, homogeneity, entropy, variance)	6
Total	**-**	**-**	**-**	**42**

3.5.2 Local Binary Pattern Features

Local Binary Patterns (LBP) were extracted to complement GLCM features and quantify local micro-textures which are critical for distinguishing subtle phenotype variations. For each pixel, LBP compares the intensity of the center pixel g_c with its *P* neighbors g_p on a circle of radius *R*:

$$LBP_{P,R}(x_c, y_c) = \sum_{p=0}^{P-1} s(g_P - g_c) \times 2^P \quad (3.10)$$

where g_C is the centre pixel value, g_P are the P neighbouring values on a circle of radius R, and

$$s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The LBP algorithm was applied using a radius of 2 pixels and 16 sampling points, employing the "uniform" method to generate rotation-invariant texture descriptors. For each pixel, the LBP code was computed by comparing its intensity to those of its 16 neighbors, assigning a binary value (0 or 1) based on whether the neighbor's intensity was less than or greater than the central pixel's intensity (Equation 3.10). The resulting binary sequence was converted to a decimal code, with uniform patterns grouped to reduce dimensionality

From these histograms, four additional metrics were derived:

1. LBP Energy: Measures histogram uniformity

$$LBP\ Energy = \sum_{k=1}^{20} h(k) \quad (3.11)$$

where h(k) is the normalized histogram in bin k

2. LBP Contrast: Measures local intensity variation within the LBP neighbourhood

$$LBP\ Contrast = \sum_{k=1}^{20} |k - \mu_{LBP}|^2 h(k) \quad (3.12)$$

where $\mu_{LBP} = \sum_{k=1}^{20} k \cdot h(k)$ is the mean LBP code value. This metric captures the variance in local texture patterns, with higher values indicating greater local intensity differences.

3. LBP Entropy: Quantifies textural complexity

$$LBP\ Entropy = \sum_{k=1}^{20} h(k) \log_2 h(k) \quad (3.13)$$

4. LBP Homogeneity: Assesses the concentration of LBP codes around the mean

$$LBP\ Homogeneity = \sum_{k=1}^{20} \frac{h(k)}{1+|k-\mu_{LBP}|} \quad (3.14)$$

These metrics, combined with the 16-bin histogram, resulted in **20 LBP descriptors**, which enhanced the ability to capture local textural differences between cording and non-cording phenotypes.

3.5.3 Regional Intensity Statistics

Basic intensity and morphological statistics were calculated for each segmented region to complement the textural features. These include:

1. Mean Intensity: The average pixel intensity within the region

$$Mean\ Intensity = \frac{1}{N} \sum_{i=0}^{N} I_i, \quad (3.15)$$

where I_i is the intensity of pixel *i* and *N* is the number of pixels in the region.

2. Intensity Standard Deviation: The spread of intensity values

$$Standard\ Deviation = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(I_i - \mu)^2}, \quad (3.16)$$

where μ_i is the mean intensity.

3. Region Area: The number of foreground pixels in the segmented region.

4. Intensity Skewness: Measures the asymmetry of the intensity distribution

$$Skewness = \frac{\frac{1}{N}\sum_{i=1}^{N}(I_i-\mu)^3}{\left(\sqrt{\frac{1}{N}\sum_{i=1}^{N}(I_i-\mu)}\right)^3}, \quad (3.17)$$

5. Intensity Kurtosis: Quantifies the tailedness of the intensity distribution

$$Kurtosis = \frac{\frac{1}{N}\sum_{i=1}^{N}(I_i-\mu)^4}{\left(\sqrt{\frac{1}{N}\sum_{i=1}^{N}(I_i-\mu)}\right)^2} - 3, \quad (3.18)$$

These descriptors provided insights into the brightness distribution characteristics of aggregate regions, enhancing the feature set.

3.5.4 Batch Processing and Data Aggregation

The feature extraction pipeline was executed in batch mode using Python (version 3.8.0) with libraries such as scikit-image (version 0.18.3) and pandas (version 1.3.4). Features were extracted from segmentation masks for all 183 images across 17 folders (15 experimental and 2 control). The resulting features were stored in a structured `dataframe`, with each row corresponding to a single image and columns representing the 42 GLCM descriptors, 20 LBP descriptors, and 5 regional statistics, totaling **67 features per image**.

3.5.5 Design Rationale

The integration of GLCM and LBP features in the texture analysis pipeline was motivated by their complementary strengths in capturing distinct aspects of textural patterns in micrscope images. This combination enhanced the ability to differentiate cording and non-cording phenotypes by providing a more comprehensive characterization of both global and local texture properties.

GLCM Characteristics: GLCM features quantify the spatial relationships between pixel intensities in an image, capturing second-order statistical properties (Equation 3.4). Haralick features, such as contrast, homogeneity, and correlation, were derived to describe global textural properties, such as the overall roughness, uniformity, or linear dependency of intensity patterns. GLCM was particularly effective for capturing large-scale structural patterns, such as the elongated, rope-like formations characteristic of cording phenotypes, across 3, 4, 5, and 7 pixel distances. However, GLCM features have limitations in capturing fine-scale, localized texture variations. The computational complexity of GLCM increases with the number of gray levels. Higher quantization levels (64 in this case) tend to reduce sensitivity to subtle micro-textures of non-cording phenotypes. Additionally, GLCM focuses on pairwise pixel relationships, which may not fully represent the intricate, localized patterns within densely-packed, disorganized aggregate regions of non-cording phenotypes.

LBP Characteristics: LBP features address these limitations by focusing on local micro-textures through a pixel-wise comparison approach. Uniform LBP patterns, with at most two bitwise transitions, were used to reduce dimensionality while preserving rotation-invariant texture information. Normalized histograms of LBP codes, along with derived metrics like energy and entropy, quantified local textural uniformity and complexity. LBP excels at capturing fine-grained, localized patterns, such as the micro-textural differences between disorganized, granular aggregates in non-cording phenotypes. Unlike GLCM, LBP is more resistant to illumination changes due to its reliance on relative intensity comparisons, making it well-suited for fluorescence microscopy images with uneven lighting artifacts.

Region Intensity Statistics: While GLCM and LBP capture spatial and textural patterns, region-based intensity statistics provide complementary information about the overall intensity distribution within each segmented area. This is relevant in the context of Mtb image analysis, where intensity may correlate with biologically meaningful properties, such as aggregate density.

3.6 Statistical Analysis and Classification Framework

The statistical analysis framework was designed to evaluate phenotypic differences captured by the 67 extracted features while establishing a classification pipeline. This section details the comprehensive methodology for normality assessment, non-parametric hypothesis testing, multiple test corrections, and effect size quantification, ensuring both statistical validity and biological interpretability.

3.6.1 Data Specification

The statistical analysis and classifier development exclusively utilized images from the cording (Parental_Phigh) and non-cording (Parental_Pminus) control groups, representing prototypical morphologies. 20 samples were randomly selected from each class to maintain statistical power and ensure class balance. The 135 unlabelled images (15 folders × 9 images) were deliberately excluded from the statistical analysis and classifier training phase for several methodological reasons:

- Phenotypic Certainty: The unlabelled may exhibit intermediate or mixed morphologies which may complicate clear class assignments
- Methodological Purity: Maintaining a clear distinction between training (known phenotypes) and application (unknown phenotypes) domains
- Preventing potential data leakage and ensuring unbiased performance estimation

This analytical approach ensured that the identified texture signatures and developed classification models reflected genuine biological differences between well-characterized cording extremes, which would have higher prediction probabilities (≥0.7) and ambiguous intermediate states which would have average prediction probabilities (±0.5). The subsequent application of the trained classifier to unlabelled samples then tested the generalizability of these learned discriminative patterns to more heterogeneous scenarios.

3.6.2 Statistical Analysis

A. NORMALITY ASSESSMENT

The statistical validity of feature comparisons between cording and non-cording classes was predicated on distributional characteristics. Normality was assessed using the Shapiro–Wilk test (Shapiro & Wilk, 1965), which is particularly well suited for moderate sample sizes (n = 20 samples) and sensitive to deviations across the entire distribution. A significance threshold of $\alpha = 0.05$ was adopted.

The results indicated that only 7 out of the 67 features (10.4%) exhibited normal distributions across both phenotypic groups (cording and non-cording) (Appendix E), while 89.6% demonstrated significant deviations from normality in at least one group. Consequently, parametric tests were deemed inappropriate for most features, necessitating the use of non-parametric statistical methods.

B. HYPOTHESIS TESTING FRAMEWORK

Given the dominance of non-normal distributions, the Mann-Whitney U test (Mann & Whitney, 1947) was employed for all phenotype comparisons. This non-parametric approach evaluates whether two independent samples originate from the same distribution, making no assumptions about underlying distributional parameters. A two-sided test was conducted while maintaining a significance threshold of $\alpha = 0.05$.

C. MULTIPLE TESTING CORRECTIONS

To address the risk of false positives arising from the simultaneous evaluation of 67 texture features, raw p-values from the Mann-Whitney U tests were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure

(Benjamini & Hochberg, 1995). This method controlled the expected proportion of false discoveries among significant results, enhancing the statistical reproducibility of the findings. The procedure was implemented as follows:

1. Rank the raw p-values in ascending order: $p_{(i)} \leq p_{(2)} \leq \cdots \leq p_{(67)}$
2. Compute critical values for each rank: $c_i = \frac{i}{67} \times 0.05$, where i is the rank and 0.05 is the target FDR level.
3. Identify the largest rank k where $p_{(k)} \leq c_k$, and declare all features with $p_{(i)} \leq p_{(k)}$ as significant.

Features with FDR-adjusted p-values less than 0.05 were deemed statistically significant, ensuring that the expected proportion of false positives among significant features remained below 5%. This approach balanced statistical power and error control, making it particularly suitable for the high-dimensional feature set used to discriminate cording and non-cording Mtb phenotypes.

D. EFFECT SIZE QUANTIFICATION

Statistical significance was complemented by effect size (practical relevance) estimation using rank-biserial correlation, calculated as:

$$r_{rb} = 1 - \frac{2U}{n_1 n_2}, \tag{3.19}$$

where U represents the Mann-Whitney U statistic, and n_1, n_2 denote sample sizes for the two groups. This correlation coefficient provides an intuitive measure of effect magnitude, with values ranging from -1 to 1, where:

- Positive values indicate higher ranks in the second group (non-cording dominance)
- Negative values indicate higher ranks in the first group (cording dominance)
- Absolute values quantify effect strength (0.1: small, 0.3: medium, 0.5: large)

E. IMPORTANCE RANKING

A composite importance score was defined to integrate FDR-adjusted significance significance and effect magnitude into a unified feature importance measure:

$$Importance\ Score = -\log_{10}(pFDR + \epsilon) \times |r_{rb}|, \tag{3.20}$$

where $\varepsilon = 10^{-10}$ ensures numerical stability. Features with low FDR-adjusted p-values and large effect sizes receive higher scores, enabling systematic ranking according to their discriminative power between cording and non-cording phenotypes. The incorporation of rank-biserial correlation with FDR-adjusted p-values addresses the critical distinction between statistical significance and practical significance in high-dimensional feature spaces. This dual-metric approach prevents over-interpretation of statistically significant but biologically trivial differences.

F. FRAMEWORK RATIONALE

The chosen statistical framework was designed to ensure interpretability in feature-level comparisons between Mtb phenotypes exhibiting cording and those exhibiting non-cording. Given the predominantly non-normal distribution of texture features, the adoption of the Mann–Whitney U test provided a non-parametric test independent of distribution, thereby allowing comparisons of central tendencies without assumptions about variance or normality. Multi-feature analysis can also result in false positives, i.e. Features erroneously having p-values $\alpha < 0.05$ meaning that the feature has high discriminative power when it actually does not have discriminative power. A multiple correction is therefore necessary. The Benjamini–Hochberg False FDR correction was applied to ensure reliable identification of genuinely discriminative features while maintaining statistical power. Complementing significance testing, rank-biserial correlation was introduced to quantify the magnitude and direction of observed effects, addressing the limitation that statistically significant results may not always be biologically meaningful. Finally, by integrating FDR-adjusted p-values and effect sizes into a unified importance score, the analysis framework prioritized features that were not statistically significant and practically relevant. This enables a biologically interpretable characterization of Mtb phenotype texture patterns.

3.6.3 Classification with Support Vector Machines (SVM)

A. SVM IMPLEMENTATION

The SVM classifier was implemented using the `scikit-learn` library (version 0.24.2) in a Python 3.8 environment. The model was initialized with the SVC class, setting `probability=True` to enable probability estimates for confidence scoring. To normalize the 67-dimensional feature space, features were standardized using StandardScaler from scikit-learn, which transformed the data to have a mean of 0 and a standard deviation of 1 based on the training set. Hyperparameter tuning was performed using `GridSearchCV` to optimize the model's performance. The parameter grid included C values of 0.1, 1, 10, and 100 (controlling the regularization strength), gamma values of `'scale'`, `'auto'`, 0.1, 0.01, and 0.001 (defining the kernel coefficient), and kernels `'rbf'` (radial basis function) and `'linear'`. The search utilized 5-fold cross-validation with `n_jobs=-1` to parallelize computation, ensuring efficient exploration of the hyperparameter space. The best model parameters were selected based on the highest cross-validation accuracy.

B. TRAINING AND TESTING

The dataset, comprising 40 labelled samples (20 cording, 20 non-cording), was split into training and test sets using `scikit-learn`'s `train_test_split` function. A 70/30 split was applied, resulting in 28 training samples and 12 test samples, with `random_state=42` for reproducibility and `stratify=y` to maintain class balance. The training process involved fitting the SVM model with the standardized training features and corresponding labels (1 for cording, 0 for non-cording). Five-fold cross-validation was conducted during hyperparameter tuning to assess model stability, with each fold preserving the class distribution. The test set was used to evaluate the final model's performance on unseen data, while the remaining 135 unlabelled samples were reserved for post-training prediction to assess generalizability.

C. EVALUATION METRICS

Model performance was evaluated using a suite of metrics calculated with `scikit-learn`'s metrics module. Accuracy was computed as the proportion of correctly classified samples, providing an overall measure of performance. Precision, recall, and F1-score were calculated for each class (non-cording and cording) to assess the model's ability to avoid false positives and detect true positives, respectively, with results aggregated using macro and weighted averages.

These were calculated as:

1. Accuracy: measured the proportion of correctly classified samples across both classes

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}, \qquad (3.21)$$

   where TP = True Positives (correctly identified cording), TN = True Negatives (correctly identified non-cording), FP = False Positives, and FN = False Negatives. This provided an overall performance measure, given as a percentage.

2. Precision: quantified the proportion of positive predictions that were actually positive which emphasizes the model's ability to avoid false positives

$$Precision = \frac{TP}{TP+FP} \qquad (3.22)$$

3. Recall (Sensitivity): measured the proportion of actual positives correctly identified, thereby assessing the model's sensitivity to cording phenotypes

$$Recall = \frac{TP}{TP+FN} \qquad (3.23)$$

4. F1-Score: provided a balanced measure of precision and recall

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (3.24)$$

5. Specificity: evaluated the model's ability to correctly identify non-cording phenotypes, complementing recall for comprehensive class evaluation

$$Specifity = \frac{TN}{TN + FP} \quad (3.25)$$

The confusion matrix was generated to detail true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), offering insights into class-specific errors. Additionally, prediction probabilities were obtained using the `predict_proba` method, enabling confidence analysis.

D. DESIGN HYPOTHESIS

The SVM was selected as the classification method based on its proven efficiency in handling high-dimensional data (Alva, et al., 2013). A hypothesis was formed that the 67 texture features could distinguish cording from non-cording phenotypes with an accuracy >90%. This hypothesis was informed by the distinct texture signatures identified in Chapter 4 (e.g., lower contrast in cording vs. higher LBP energy in non-cording) and the need for a model capable of managing correlated features without requiring extensive computational resources. The features showed high collinearity. However, a strategic decision was then made to all features despite high collinearity. The rationale for this decision was that retaining all 67 features preserved the full texture spectrum, with an expectation that hyperparameter tuning and cross-validation would reduce overfitting risks. This resulted in a high-dimensional feature which could be best handled by an SVM.

Table 3.3: A comparative analysis between SVM and Random Forest revealed that the SVM was more suited to the classification task

Feature	**Support Vector Machine (SVM)**	**Random Forest (RF)**	**Rationale for SVM Selection**
High-Dimensional Performance	Excellent with regularization	Good, but cannot be overfit with many features	SVM's maximum margin principle better handles `n_features` > `n_samples`
Feature Correlation Handling	Robust using kernel trick	Can be sensitive to correlated features	Linear SVM demonstrates resilience to multicollinearity identified in our feature set
Interpretability	Clear decision boundaries	Feature importance scores	SVM offers transparent geometric boundaries crucial for biological interpretation
Computational Efficiency	Efficient with linear kernel	More computationally intensive	Faster training and prediction for our dataset size
Class Imbalance Handling	Good with class weights	Native handling through sampling	Both adequate; SVM's explicit regularization preferred

3.7 Computational Environment

This section outlines the computational infrastructure and software tools utilized to develop, train, and evaluate the automated image processing pipeline for Mtb phenotype classification. The computational environment was designed to ensure performance, scalability, and reproducibility across all stages of the pipeline.

3.7.1 Hardware and Software Specifications

All experiments were conducted on **Google Colaboratory (Colab)**, a cloud-based computational platform providing on-demand access to GPU acceleration. The environment was configured with a single **NVIDIA T4 GPU** (16 GB VRAM), **2 virtual CPU**s, and approximately **12 GB of system RAM**. This setup provided sufficient computational capacity for training the U-Net segmentation model, extracting texture features, and executing classification procedures on the dataset. The cloud-based configuration eliminated the need for local high-performance hardware while supporting reproducible experimental workflows through notebook-based development.

The software stack was based on **Python 3.8**, integrating several specialized libraries for different components of the pipeline. `TensorFlow 2.5` was employed for U-Net implementation and training, leveraging GPU acceleration for efficient processing of 1024×1024 image tiles. `scikit-learn 0.24.2` was used for SVM classification, feature scaling, and model evaluation, while `scikit-image 0.19.1` facilitated the computation of GLCM and LBP texture features. `OpenCV 4.5` was used for image pre-processing, including grayscale conversion, edge detection, and the generation of weak labels through the Edge-Dilated Weak Masking (EDWM) strategy. All analyses were conducted in **Jupyter Notebook** environments on **Colab**, enabling structured development and visualization.

3.7.2 Efficiency Considerations

To optimize computational resource utilization, multiple efficiency measures were implemented throughout the image processing and analysis pipeline. During U-Net training on 288 image tiles, batch processing with a batch size of 2 was employed. This configuration reduced GPU memory consumption and enabled processing of 1024 × 1024 pixel tiles within the 16 GB memory limit of the NVIDIA T4 GPU. The average training time was approximately **1 hour for 100 epochs**, with validation performed after each epoch to monitor convergence and prevent overfitting.

Feature extraction, including the computation of GLCM and LBP features across 175 segmented samples, was parallelized using `NumPy` **vectorization** and `scikit-image's` **multi-threading capabilities**. This parallelization reduced the total processing time to approximately **0.2 hours**, despite the computationally intensive nature of multi-scale texture calculations.

Disk storage requirements were managed through efficient intermediate data handling, which included the use of compressed file formats (.csv) for storing feature matrices, and model outputs. This strategy minimized the total disk footprint to approximately **2 GB** for the complete pipeline.

3.8 Experimental Design and Validation Strategy

3.8.1 Overall Experimental Framework

The experimental design for the automated phenotype classification pipeline was structured to ensure a reliable and systematic evaluation process. Each pipeline component: pre-processing, segmentation, feature extraction, and classification, was individually validated before integration into the complete system. This modular approach facilitated the precise identification of performance limitations and prevented error propagation across stages, enhancing overall reliability.

The framework was guided by three principles:

- **Progressive Complexity:** Initial assessments utilized simpler techniques before advancing to more complex algorithms, allowing for a stepwise evaluation of efficiency.
- **Ablation Testing:** The contribution of each component was evaluated by systematically removing elements and determining their individual impact on performance.
- **Cross-Validation:** Reliability was ensured through data partitioning strategies, which provided a comprehensive assessment of model stability and generalizability.

3.8.2 Hypothesis Formulation and Testing

The goal of the project was to quantify texture features of Mtb cording phenotypes and use these quantifications to distinguish phenotypes exhibiting cording and those exhibiting non-cording. The central research hypothesis was therefore: "Quantitative texture features derived from low resolution images of Mtb aggregates can effectively discriminate between cording and non-cording phenotypes with a classification accuracy >90%." This hypothesis was broken down into three testable sub-hypotheses:

- H_1: Multi-scale texture features demonstrate statistically significant differences between cording and non-cording phenotypes and can be evaluated using the Mann-Whitney U test with a significance level of $\alpha = 0.05$.
- H_2: Weakly-supervised segmentation achieves Dice coefficients greater than 0.60 while preserving biological relevance, assessed through segmentation metrics.
- H_3: The integrated pipeline generalizes to unseen data with a classification accuracy of at least 90%, validated through hold-out testing.

The design aimed to validate this hypothesis through a controlled experimental setup and comprehensive performance evaluation.

3.8.3 Variable Definition and Control

Clearly defining independent, dependent, and controlled variables ensured that the research was conducted systematically, allowing for the isolation of the pipeline's effects on phenotype classification. To ensure reliable evaluation, variables were defined and controlled as follows

Independent Variables:

- **Image Pre-Processing Parameters** (e.g., filter sizes): These settings determine the quality of input images, directly affecting subsequent segmentation and feature extraction accuracy.
- **U-Net Architectural Configurations** (e.g., depth, filter settings): Variations in network design influence segmentation performance
- **Feature Extraction Parameters** (e.g., GLCM distances, LBP configurations): These settings shape the texture features' discriminative power, critical for distinguishing cording from non-cording phenotypes.
- **SVM Hyperparameters** (e.g., kernel type, regularization strength): These control the classifier's learning behaviour, impacting its ability to generalize to unseen data.

Dependent Variables:

- **Segmentation Performance Metrics** (Dice coefficient, Intersection-over-Union): These quantify the accuracy of aggregate region identification, which is the U-Net's efficiency.
- **Statistical Significance Indicators** (p-values, effect sizes): These assess the reliability of texture feature differences between phenotypes, supporting hypothesis testing.
- **Classification Performance Metric**s (accuracy, precision, recall, F1-score): These measure the SVM's success in phenotype prediction, crucial for evaluating the pipeline's overall performance.

Controlled Variables:

- **Consistent Image Acquisition Size** (20× magnification, 2592×1944 resolution): Standardized imaging ensures uniformity, minimizing variability.
- **Fixed Control Phenotypes** (Parental_Phigh for cording and Parental_Pminus for non-cording): Using defined control groups provides a reliable baseline for training and validation.

3.8.4 Validation Hierarchy

A tiered validation strategy was implemented as follows to assess both model stability and generalizability:

Tier 1: Component-Level Validation:

- Segmentation: Utilized an 80/20 train-validation split with stratified sampling to maintain class balance.
- Feature Extraction: Validated feature distributions statistically to confirm discriminative power.
- Classification: Employed nested cross-validation with hyperparameter tuning to optimize model performance.

Tier 2: Integration Testing:

- Compared classification results against manual annotations by domain experts for accuracy.

Tier 3: Generalization Assessment:

- Performed hold-out testing on 135 unseen samples to evaluate real-world applicability.

3.8.5 Performance Metrics

A comprehensive set of performance metrics was selected to evaluate the pipeline's effectiveness across segmentation and classification stages. For U-Net segmentation, the Dice coefficient and Intersection-over-Union (IoU) were computed to measure overlap between predicted and ground-truth masks. These metrics were chosen for their sensitivity to segmentation accuracy in medical imaging. For SVM classification, accuracy (overall correctness), precision (positive predictive value), recall (sensitivity), and F1-score (harmonic mean of precision and recall) were calculated, with results reported for both test and unseen datasets. The confusion matrix provided a breakdown of true positives, true negatives, false positives, and false negatives for all samples, while prediction probabilities enabled confidence analysis. These metrics were selected to align with the hypothesis, ensuring a thorough assessment of phenotype discrimination.

3.8.5 Application Overview

The experimental design and validation strategy outlined in this section were systematically applied across the pipeline. The Progressive Complexity principle guided the development in Sections 3.3 (Pre-processing) and 3.4.2 (Weak Label Generation), where simpler techniques preceded advanced modelling. Ablation Testing was implemented in Section 3.4 (Segmentation) to assess component contributions, while Cross-Validation was utilized in Sections 3.4 and 3.6.6 to ensure model stability. Hypotheses H1, H2, and H3 were evaluated in Sections 4.5 (Texture Analysis), 4.3 (Segmentation), and 4.6 (Classification) respectively.

3.9 Chapter Summary

The methodology outlined above integrates automated segmentation, comprehensive multi-scale texture feature extraction, statistical analysis, and optimized classification to enable discrimination between cording and non-cording aggregates. The combination of U-Net-based segmentation and feature engineering provides a reproducible foundation for quantitative image-based phenotype analysis. The next chapter presents the results obtained using this pipeline, including segmentation performance, feature discriminability, and classification accuracy.

Chapter 4

Results

4.1 Introduction

This chapter presents the results of the proposed pipeline. The primary objective of this project was to develop an image processing pipeline that could automatically quantify phenotypes of Mycobacterial tuberculosis and use these quantifications to classify whether the image exhibits is cording or non-cording characteristics. The pipeline involved data curation, image processing, segmentation using a U-Net model, feature extraction, and classification using a Support Vector Machine (SVM). Statistical analysis was also conducted to compare features that could discriminate between cording and non-cording phenotypes.

The results are organized as follows: Section 4.2 details the data curation and pre-processing outcomes. Section 4.3 presents the segmentation performance of the U-Net model, while Section 4.4 assesses extracted phenotype features. Section 4.5 provides a quantitative analysis and statistical comparisons of phenotypes. Section 4.6 reports the performance of the SVM classifier for categorizing phenotypes into corded and non-corded. Section 4.7 summarizes the computational efficiency of the system. Visualisations, including images, tables and charts support the quantitative findings. Interpretation and evaluation of results will be performed in Chapter 5 (Discussion).

4.2 Data curation and Pre-Processing Outcomes

A dataset of 183 images was curated. Images were acquired at 20× magnification with a resolution of 2592×1944 pixels and organized into a structured directory with standardized filenames. 135 images were grouped into 15 folders (9 images each), representing different variations and 48 images were grouped into two control folders (24 images each for cording and non-cording phenotypes).

Pre-processing was performed to reduce acquisition artifacts such as noise and uneven lighting, while preserving phenotypic structures. A median filter was applied to smoothen noise without blurring the images. Root Mean Square (RMS) Contrast was used to quantify noise levels. Table 4.1 presents the average RMS Contrast and its standard deviation for raw vs filtered images across the 17 folders.

As shown in Table 4.1, the median filter consistently reduced RMS Contrast across all folders, with raw RMS values ranging from 6.67 (Ci-E06-2) to 8.49 (Ci-E06-8) and filtered RMS values from 5.80 (Ci-NT-4) to 7.44 (Ci-U17-4). The standard deviation indicated varying noise levels across folders. The average RMS reduction after filtering was 12.4%, with Ci-U17-3 (14.4%) and Ci-A04-5 (14.1%) showing the largest reductions, which suggest that these folders were more sensitive to the filter (Figure 4.1). The relatively small RMS changes (0.75 to 1.09), show that the median filter reduces noise while preserving phenotypic structure details. Figure 4.2 illustrates an image pair from the Parental_Pminus folder. The raw image had visible noise artifacts that were effectively reduced after filtering. The filtered image showed a slightly smoother background while retaining phenotype structure. This will result in a more refined mask being generated down the pipeline.

Table 4.1: Average RMS Contrast Before and After Median Filtering

Folder	RMS Raw (Avg. ± Std. Dev.)	RMS Filtered (Avg. ± Std. Dev.)
Ci-A04-1	7.87 ± 0.32	6.91 ± 0.28
Ci-A04-2	6.85 ± 0.28	5.94 ± 0.24
Ci-A04-3	7.63 ± 0.31	6.77 ± 0.27
Ci-A04-5	6.75 ± 0.27	5.80 ± 0.23
Ci-E06-1	7.69 ± 0.31	6.66 ± 0.27
Ci-E06-2	6.67 ± 0.27	5.80 ± 0.23
Ci-E06-7	7.52 ± 0.30	6.76 ± 0.27
Ci-E06-8	8.49 ± 0.34	7.39 ± 0.30
Ci-NT-1	7.64 ± 0.31	6.76 ± 0.27
Ci-NT-2	7.40 ± 0.30	6.48 ± 0.26
Ci-NT-3	6.97 ± 0.28	6.19 ± 0.25
Ci-NT-4	6.74 ± 0.27	5.80 ± 0.23
Ci-U17-2	7.14 ± 0.29	6.25 ± 0.25
Ci-U17-3	6.79 ± 0.28	5.81 ± 0.23
Ci-U17-4	8.39 ± 0.34	7.44 ± 0.30
Parental_Phigh	7.30 ± 0.29	6.47 ± 0.26
Parental_Pminus	7.57 ± 0.31	6.67 ± 0.27

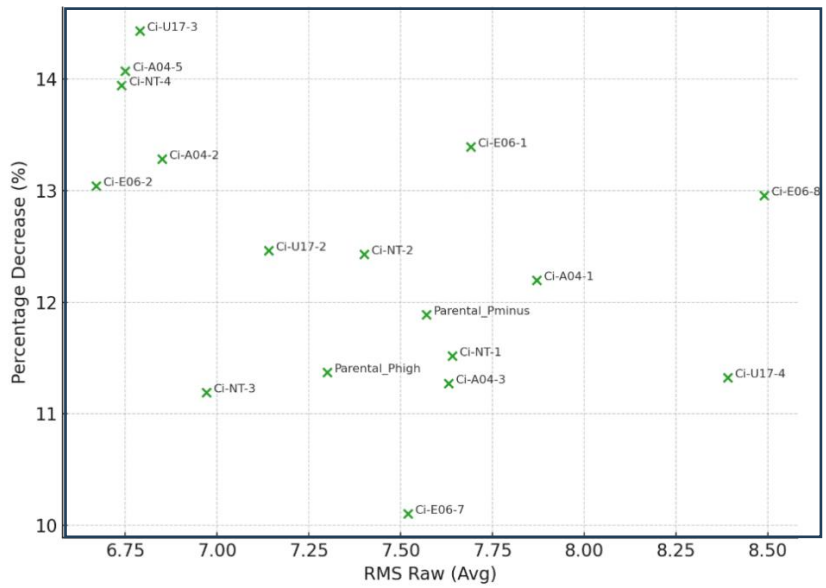
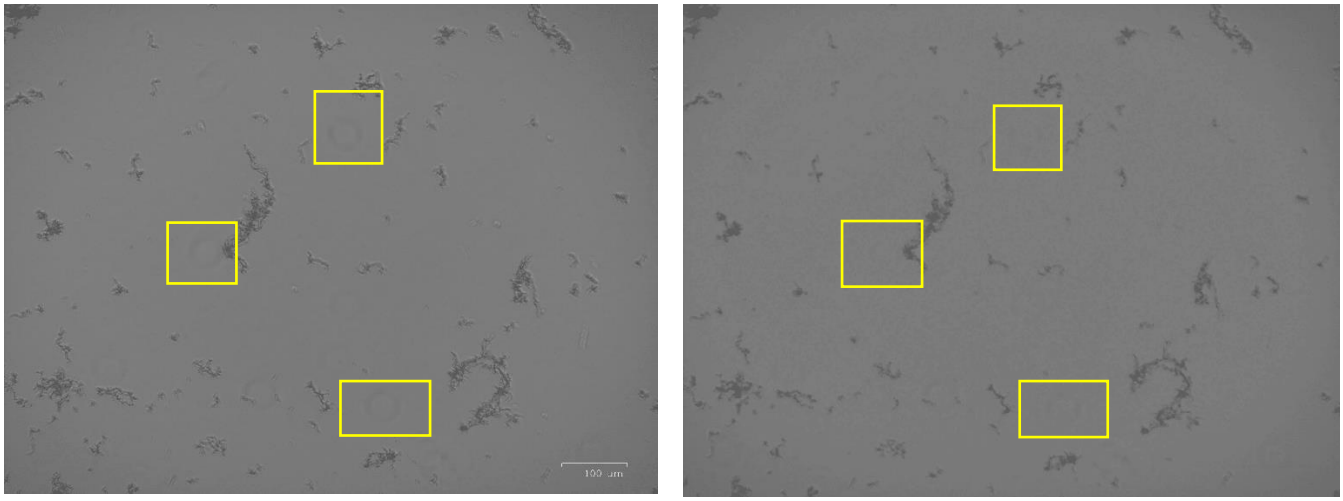


Figure 4.1: Relationship between Raw RMS and Percentage Decrease for each folder



(a) Raw image (b) Filtered image

Figure 4.2: Highlighted (yellow rectangle) regions show some of the noise artefacts that have been removed after filtering.

4.3 Image Segmentation

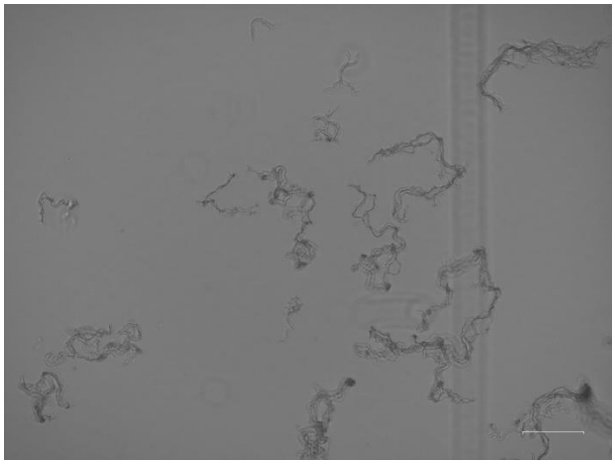
Images were segmented using a U-Net neural network. The model utilized control data as its labeled data for training: 24 images from Parental_Phigh (cording) and 24 images from Parental_Pminus (non-cording). After training and evaluation, the model was applied to the remaining folders which had unseen data of 135 images.

4.3.1 Weak Labelled Masks

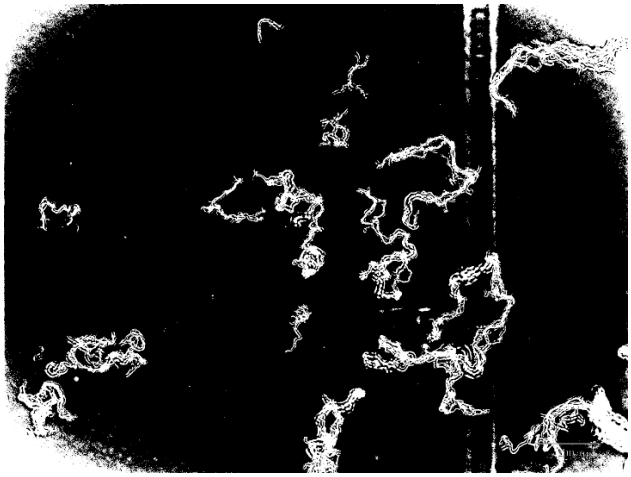
To train the U-Net model, weakly labelled masks were required. Initial attempts to generate masks using Otsu thresholding produced inconsistent masks, particularly struggling with image boundaries and background separation due to uneven illumination of raw images. To address this Edge-Dilated Weak Masking (EDWM) was used. This method generated approximate aggregate masks by applying Canny edge detection followed by morphological dilation, resulting in coarse but sufficiently informative weak masks.

Figure 4.3(a) shows a sample grayscale image exhibiting the cording phenotype. Otsu thresholding produced inconsistent masks that struggled with image edge boundaries and uneven lighting (Figure 4.3(b)). In contrast, the EDWM technique (Figure 4.3(c)) produced a more refined binary representation of the cording phenotype. By combining edge detection with controlled dilation, EDWM enhanced boundary definition while reducing noise artifacts. The resulting masks show cleaner traces of the cords while preserving morphology.

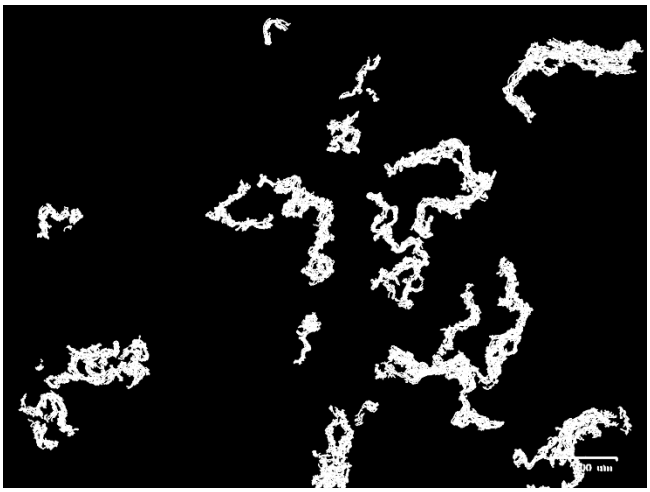
Given its ability to handle uneven illumination of raw images, the EDWM masks were selected for training the U-net segmentation model which created more refined masks.



(e) Raw image



(f) Generated image mask after applying Otsu thresholding



(c) Generated mask after applying EDWM

Figure 4.3: Comparing effect of Otsu thresholding vs EDWM to create weak masks

4.3.2 U-Net Training

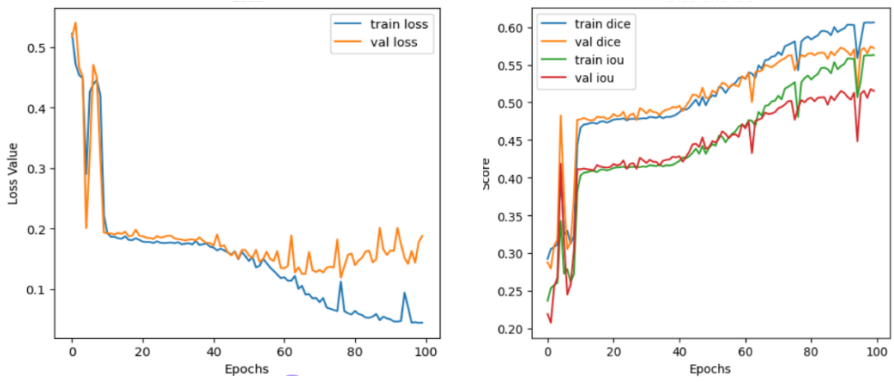
A. DATASET PREPARATION

Due to computational constraints and model input requirements, original images were partitioned into tiles of 1024×1024 pixels with 25% overlap between adjacent tiles. This pre-processing strategy generated 288 total tiles for model training and evaluation, with a class distribution of 143 tiles containing Parental_Phigh (cording) phenotype, and 144 tiles containing Parental_Pminus (non-cording) phenotype. The dataset was split into training (80%, 230 tiles) and validation (20%, 58 tiles) sets using a stratified approach to maintain class balance.

B. MODEL ARCHITECTURE AND TRAINING

A U-Net architecture was implemented with input dimensions of 1024×1024×1. The model was trained using weighted categorical cross-entropy loss with computed class weights of 0.67 for background, 1.34 for Parental_Phigh, and 1.33 for Parental_Pminus to address class imbalance.

Training was done for 100 epochs with a batch size of 2. An Adam optimizer with an initial learning rate of 1e-4 was used. As shown in Figure 4.4(a), the model demonstrated stable convergence with training loss decreasing from 0.567 to 0.050. Validation loss showed greater variability but stabilized around 0.150-0.200 after epoch 50, indicating effective regularization with minimal overfitting.



(a) (b)

Figure 4.4: (a) Loss curves showing consistent decrease in training loss and stabilized validation loss over 100 epochs, (b) Dice coefficient and IoU showing improvements in training and validation, indicating effective generalization.

C. SEGMENTATION METRIC PROGRESSION

The training curves (Figure 4.4(b)) revealed consistent improvement in segmentation performance throughout the training process. The Dice coefficient increased from 0.274 to 0.608 (training) and 0.288 to 0.572 (validation), while Intersection-over-Union improved from 0.215 to 0.565 (training) and 0.220 to 0.515 (validation). The model had 3 learning phases: rapid initial improvement (epochs 1-20), sustained learning (epochs 21-70), and convergence (epochs 71-100). After 40 epochs, both training and validation metrics progressed in parallel. This demonstrated that the model was generalizing well to unseen data.

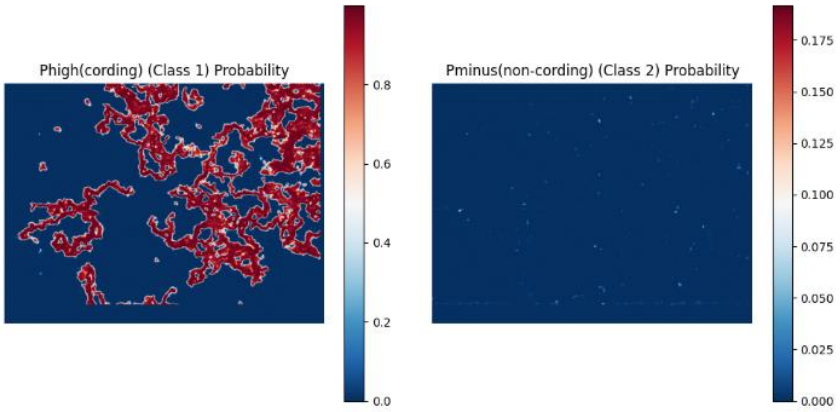
D. QUANTITATIVE SEGMENTATION PERFORMANCE

The U-Net model's performance was evaluated using class-specific and overall Dice coefficients and Intersection-over-Union (IoU) metrics, computed across the validation set after training on 230 tiles and validating on 58 tiles. The Dice coefficient measures the overlap between predicted and ground-truth masks, with higher values indicating better segmentation accuracy. Table 4.2 presents the comparative metrics for cording (Phigh) and non-cording (Pminus) images, revealing consistent performance across phenotype classes.

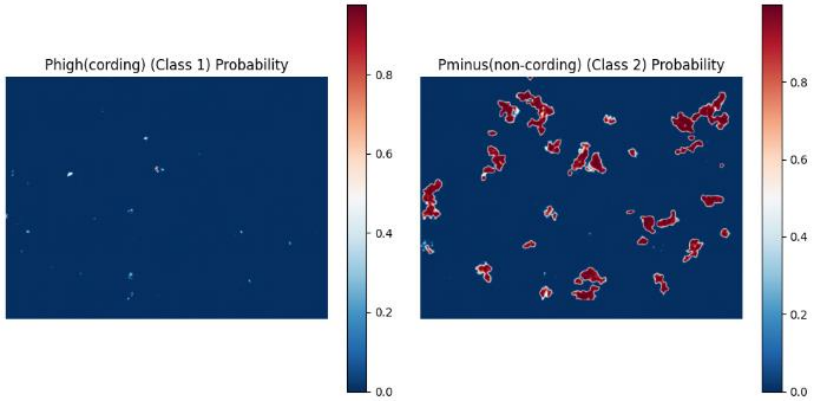
Table 4.2: Comparative analysis segmentation across phenotypic classes

Metric	Phigh Images	Pminus Images
Background Dice	0.962	0.990
Background IoU	0.927	0.980
Phigh Dice	0.887	0.000
Phigh IoU	0.796	0.000
Pminus Dice	0.000	0.851
Pminus IoU	0.000	0.740
Overall Dice	0.616	0.614
Overall IoU	0.575	0.574

From Table 4.2, the model demonstrated good background segmentation accuracy and had a strong performance on segmenting both target classes. The model also showed good class discrimination with zero cross-prediction between phenotype classes. This can be further illustrated by the probability maps of two sample images (Figure 4.5).



(a) Probability maps for cording sample



(b) Probability map for non-cording sample

Figure 4.5: Probability map analysis showing (a) Phigh image with class probability distributions, (b) Pminus image with probability maps, demonstrating well-calibrated confidence estimates and effective class separation with little to no overlap.

The probability maps show bright intensities within the core regions of target structures. The bright areas indicate high model confidence in these areas. For Phigh images, the Class 1 (cording) probability map displays strong activation along the dense, cord formations, while the Class 2 (non-cording) probabilities remain near zero throughout the image (Figure 4.5(a)). The clear separation between class probabilities demonstrates that the model learned distinctive feature representations for each phenotype:

- In Phigh images: Class 1 probabilities dominate in cording regions while Class 2 probabilities remain minimal (Figure 4.5(a))
- In Pminus images: Class 2 probabilities activate in non-cording regions while Class 1 probabilities stay low (Figure 4.5(b))

The class-specific Dice scores demonstrate the model's strong ability to segment the dominant phenotype in each image set: 0.887 for Phigh images and 0.851 for Pminus images. The background class achieved excellent Dice scores (0.962 for Phigh, 0.990 for Pminus), indicating effective separation of non-bacterial regions, which comprised the majority of each tile.

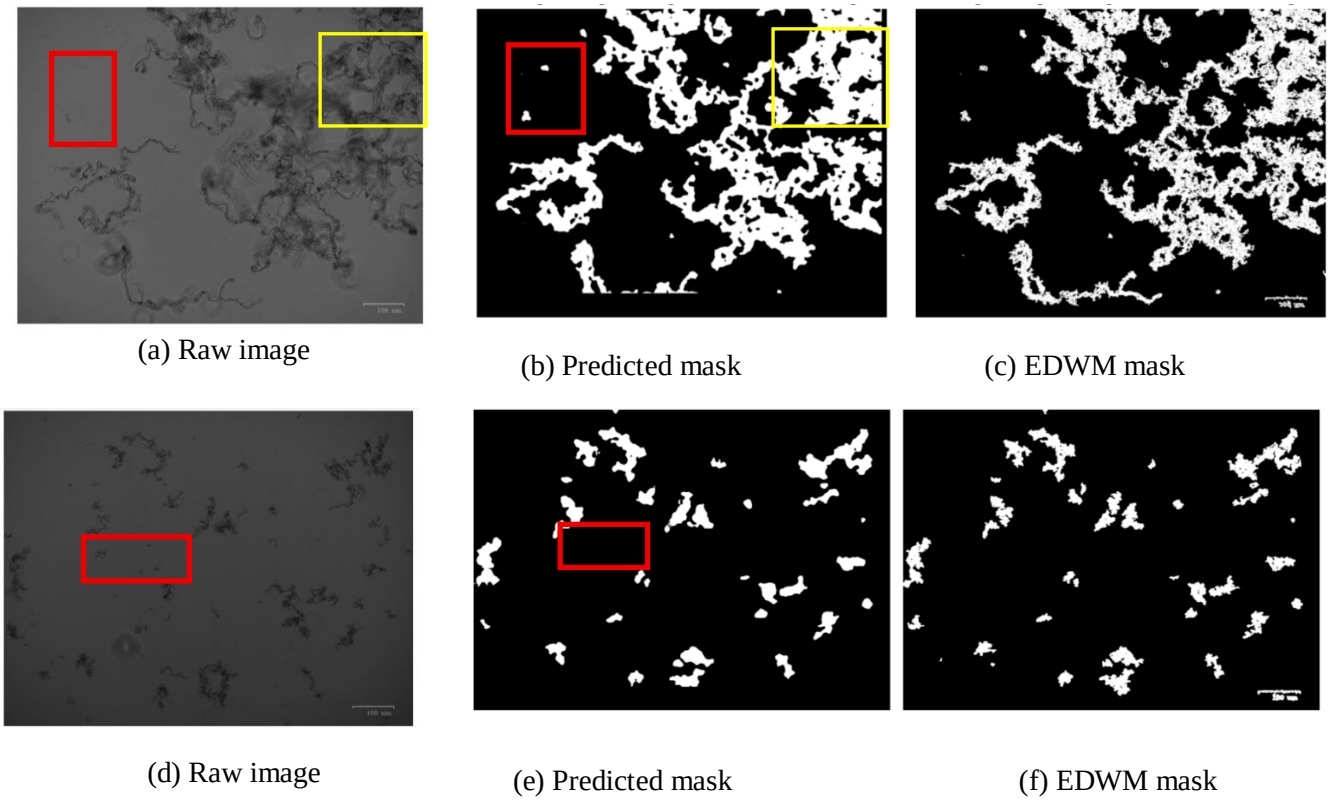
However, the overall Dice (0.616 for Phigh images, 0.614 for Pminus images) is notably lower than the class-specific scores due to the multi-class nature of the segmentation. The overall metric represents a macro-average across all three classes (background, Phigh, Pminus), incorporating zero scores for the minority phenotype in each set (e.g., Pminus Dice = 0.000 in Phigh images, as the model does not detect absent classes). This averaging dilutes the high

performance on dominant classes and background. These results mean that the model might struggle to generalize to mixed phenotype scenarios. The discrepancy underscores the effectiveness of weak supervision with EDWM for targeted phenotype segmentation but suggests limitations in handling underrepresented classes, potentially due to the balanced but small tile dataset (288 tiles total) and the absence of mixed-phenotype training images.

Corresponding IoU values follow a similar pattern, with class-specific IoU (0.796 for Phigh, 0.740 for Pminus) exceeding overall IoU (0.575/0.574), reinforcing the averaging effect. These results validate the pipeline's segmentation efficiency for phenotype-specific tasks while indicating opportunities for improvement, such as incorporating more ambiguous representations of phenotypes to improve overall performance in diverse dataset.

E. QUALITATIVE ANALYSIS

Visual analysis of the images show that the model achieved a high degree of accuracy in segmenting the target structures. For the cording phenotype (Figure 4.6(a-c)), the network-like patterns were accurately traced. Similarly, for the non-cording phenotype (Figure 4.6(d-f)), sparse aggregates were delineated with an acceptable level of accuracy. The predicted masks exhibited smooth, continuous boundaries and showed consistent performance across varying structural densities.



(a) Raw image (b) Predicted mask (c) EDWM mask

(d) Raw image (e) Predicted mask (f) EDWM mask

Figure 4.6: Mask refinement demonstration: EDWM training masks vs. final model predictions showing improved boundary precision and class discrimination.

There is slight under-segmentation which is evidenced by some parts of clumped cords getting erroneously merged in Figure 4.6(b), indicated by yellow rectangles. There are also small, isolated false positives and false negatives that can be observed in both generated masks, indicated by red rectangles. However, these errors are minimal and do not affect the overall morphological shape of the segment.

F. CONCLUSION

The results confirm that the U-net model is highly effective for identifying and segmenting cording and non-cording phenotypes. This was shown by stable convergence throughout training with minimal overfitting which indicated good generalization despite noise in the weak labels. The model was able to correctly distinguish between cording and non-cording morphologies while maintaining well-calibrated probability estimates. These results validate the effectiveness of weak supervision for segmenting cording phenotypes.

4.4 Texture Feature Extraction using Gray Level Co-occurrence Matrix

4.4.1 Overview

Texture analysis was performed on segmented bacterial regions from all images using Gray Level Co-occurrence Matrix (GLCM) methodology. The analysis focused on pixels identified as bacterial aggregates through U-Net segmentation, eliminating background interference. The feature extraction process employed a multi-scale GLCM approach to characterize phenotypic textures across spatial relationships.

4.4.2 Multi-Scale GLCM Feature Extraction

Multiple spatial scales were used to capture phenotype patterns across different structural granularities: Medium-scale textures (distances of 3-5pixels) characterized fine structural patterns, while macro-scale textures (7pixel distance) captured broader organizational features. For each scale, six statistical properties (contrast, dissimilarity, homogeneity, energy, correlation, and angular second moment) were computed across multiple orientations (0° and 45°), generating 42 comprehensive texture descriptors that quantified both individual spatial relationships and aggregate statistical distributions across scales.

4.4.3 Feature Extraction Output and Data Quality

The statistical analysis of texture features revealed significant variations across the bacterial phenotypes. Aggregate pixel counts averaged 527,953 ± 345,237, with a coverage ratio of 10.64% ± 6.98 which indicated that on average, there was substantial coverage of aggregate in each image. Medium-scale contrast averaged 36.7557 ± 19.1814 which showed variability in fine structures, while macro-scale contrast of 59.9452 ± 31.0632, suggesting greater intensity variations at border regions. Homogeneity remained high at both medium (0.9115 ± 0.0553) and macro (0.9002 ± 0.0614) scales which showed that there were consistent texture patterns with some variability. The coverage distribution ensured representation of the diverse phenotype structures for texture analysis. The key statistics are summarized in Table 4.3:

Table 4.3: Summary Statistics of Key LBP Features Across All Images.

Feature Category	Avg. ± Std. Dev.	Min.	Max.
1. Aggregate Coverage			
Aggregate Pixels	535,923 ± 351,765	35,636	1,939,977
Coverage Ratio	10.64% ± 6.98%	0.71%	38.50%
2. Medium-scale Features			
Contrast	36.7557 ± 19.1814	4.6977	97.6693
Homogeneity	0.9115 ± 0.0553	0.6906	0.9934
3. Macro-scale Features			
Contrast	59.9452 ± 31.0632	7.6141	157.5852

Homogeneity	0.9002 ± 0.0614	0.6587	0.9922

The complete feature matrix with 175 samples × 67 features provides enough data for further statistical analysis and phenotypic classification.

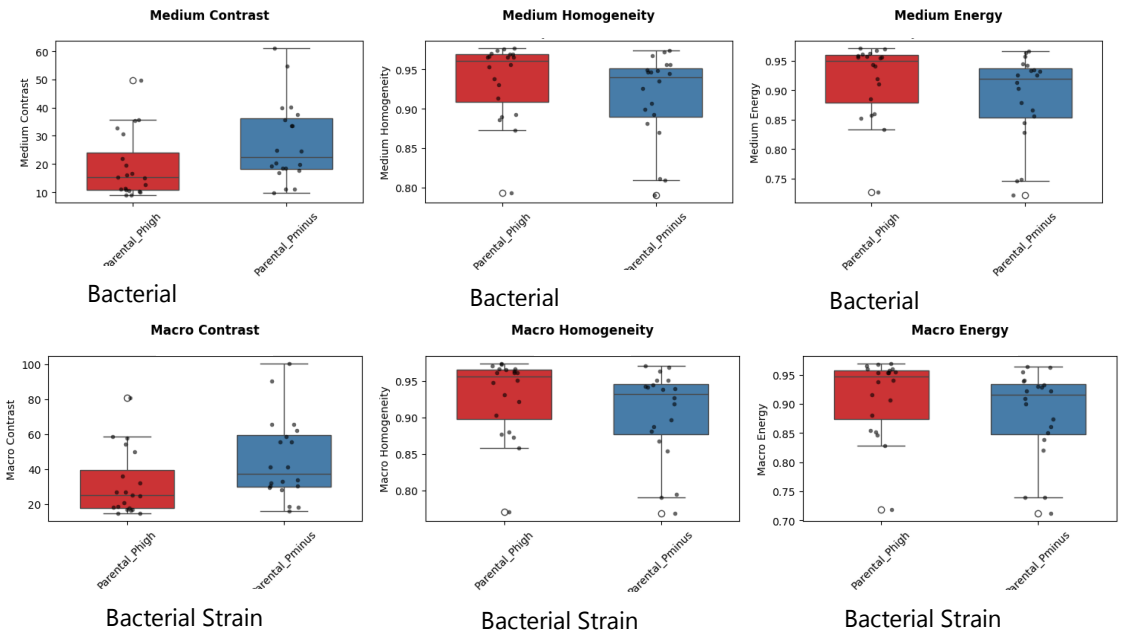
4.5 Statistical Analysis of Texture Features for Phenotype Classification

Texture analysis revealed distinct phenotypic signatures between cording (Parental_Phigh) and non-cording (Parental_Pminus) phenotypes across multiple spatial scales. Analysis of 20 samples per phenotype demonstrated several statistically significant differences with substantial effect sizes.

4.5.1 Phenotype Texture Signatures

The multi-scale texture analysis revealed distinct profiles between the Parental_Phigh (cording) and Parental_Pminus (non-cording) phenotypes, highlighting differences in structural organization and intensity variations. For Parental_Phigh, medium-scale features showed lower contrast (19.1758 ± 11.2385) and higher homogeneity (0.9361 ± 0.0468) and energy (0.9171 ± 0.0617), suggesting smoother, more uniform fine-scale patterns consistent with cording structures. At the macro scale, contrast was moderate (31.4093 ± 18.3276), with high homogeneity (0.9285 ± 0.0519) and energy (0.9136 ± 0.0637), which showed more organized broader arrangements. LBP energy was lower (0.1361 ± 0.0066), indicating less local texture complexity, while region standard intensity deviation (12.7365 ± 2.0479) and bacteria pixels (392,799.6 ± 296,798.5) showed moderate variability and coverage.

In contrast, Parental_Pminus exhibited higher medium-scale contrast (27.4145 ± 13.8786) with slightly lower homogeneity (0.9141 ± 0.0548) and energy (0.8865 ± 0.0731), which higher intensity which indicates irregular, scattered aggregates. Macro-scale contrast was notably higher (45.2361 ± 22.6855), with lower homogeneity (0.9046 ± 0.0600) and energy (0.8819 ± 0.0753). This suggests that there was greater variability in larger structural features. LBP energy was significantly higher (0.1562 ± 0.0080), reflecting increased local pattern diversity, while region standard intensity deviation (11.9472 ± 1.5191) was slightly lower, and bacteria pixels (539,297.7 ± 352,347.8) indicated broader coverage. Box plots were utilized to visualize the distribution of these features:



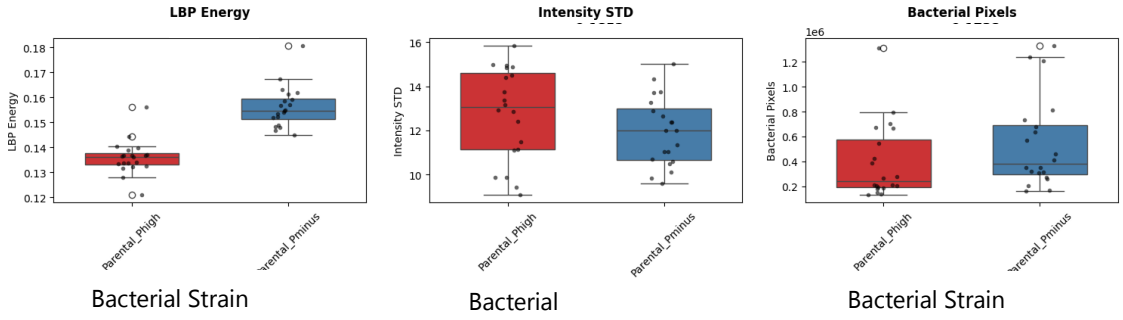


Figure 4.7: Phenotypic signatures of cording (red) and non-cording (blue) characterized using box plots.

The central line within each box represents the median feature value, while the box boundaries correspond to the interquartile range (IQR), encompassing the middle 50% of the data. The whiskers extend to the most extreme data points within 1.5×IQR from the lower and upper quartiles, thereby indicating the general spread of the observations. Individual data points are plotted to visualize the sample distribution within each group (n =20 per phenotype). Filled circles represent data within the whisker range, whereas open circles represent outliers, defined as values falling outside 1.5×IQR from the box limits. The p-values above each plot correspond to the results of the statistical comparison (Mann–Whitney U test) between phenotypes.

These signatures suggest that Parental_Pminus (non-cording) phenotypes have sharper, more variable textures, likely due to their disorganized, disaggregated growth, whilst Parental_Phigh (cording) phenotypes display smoother, more homogeneous patterns, likely from their slightly more organized and tightly packed rope-like structures.

4.5.2 Statistical Discrimination of Phenotypes

The initial step in the statistical analysis involved assessing the normality of the 67 extracted texture features using the Shapiro-Wilk test to determine the appropriate parametric or non-parametric test for comparing Parental_Phigh and Parental_Pminus phenotypes (See Appendix E1). The results revealed that only 7 features (10.4%) were normally distributed in both groups, while 33 features (49.3%) were non-normal in both groups, with an additional 27 features (40.3%) showing normality in only one group. This means up to 89% of features did not show normality in both phenotypes.

4.5.3 Normality Test

Given the predominance of non-normal distributions, the Mann–Whitney U test was applied to assess statistical differences in texture features between the two phenotypes. This test, which does not assume distributional normality, provided comparison of median feature values between groups. For each feature, the corresponding rank-biserial correlation coefficient r_{rb} was computed as a measure of effect size, directly derived from the U statistic. The coefficient quantifies both the strength and direction of group differences, with negative values indicating higher feature values in the cording group and positive values indicating dominance in the non-cording group.

4.5.4 Multiple Testing Correction and Feature Significance

To control for the inflation of false positives resulting from the simultaneous testing of 67 features, p-values from the Mann–Whitney U tests were adjusted using the Benjamini–Hochberg FDR procedure. This adjustment ensured that the expected proportion of false discoveries among significant features remained below 5% (FDR < 0.05). Following correction, 24 features remained statistically significant (Figure 4.9), confirming their discriminative power between cording and non-cording phenotypes.

4.5.5 Effect Size and Feature Importance Ranking

Effect sizes were further incorporated into an integrated feature ranking framework using the rank-biserial correlation and FDR-adjusted significance. Features demonstrating both low FDR-corrected p-values and large absolute effect sizes ($|r_{rb}| \geq 9$) were considered the most discriminative (Figure 4.8). Negative correlations signified features predominant in cording phenotype (red), while positive correlations denoted dominance in non-cording phenotype (blue). Figure 4.9 shows ranking of the 24 features based on the importance score rankings: cooler hues indicate features of higher importance (stronger statistical significance) and warmer hues indicate relatively lower-ranked, yet significant, features. (See full table in Appendix E2)

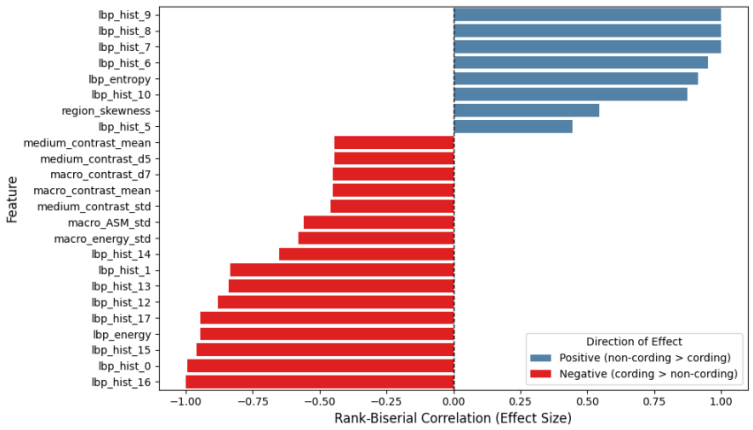


Figure 4.8: Effect sizes for FDR-Significant Features

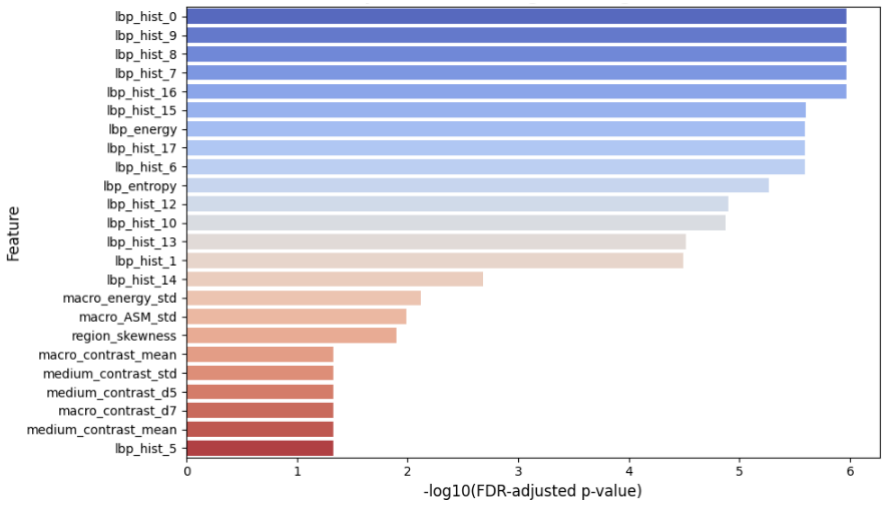


Figure 4.9: Feature Ranking Analysis (FDR Significance). Top 10 rankings were dominated by LBP features.

The feature ranking results were strongly dominated by LBP histograms. LBP histograms quantify the frequency of distinct local texture patterns across the image. These patterns can effectively encode microstructural characteristics such as smoothness, edges, and irregular patterns. Specifically:

- Lower bins (0–3) correspond to uniform patterns and homogeneous regions, reflecting smoother, structurally regular textures.
- Middle bins (7–9) represent edge- and corner-like structures, capturing transitional boundaries and organized local gradients.
- Higher bins (15–17) describe complex, non-uniform textures, indicative of irregular, heterogeneous microstructures.

The statistical dominance of middle-bin features (`lbp_hist_7-9`) in Parental_Phigh indicates a higher prevalence of organized edge and boundary textures, consistent with the cording phenotype, which is characterized by elongated, filamentous bacterial aggregate arrangements. Higher-bin features (`lbp_hist_15-17`) were more pronounced in Parental_Pminus, suggesting the presence of more disordered and heterogeneous local textures associated with non-cording aggregates.

LBP-derived energy and entropy also confirmed these findings:

- LBP Energy and LBP Histogram Uniformity were higher in Parental_Pminus, consistent with increased textural granularity of non-cording aggregates.
- LBP Entropy was higher in Parental_Phigh, reflecting greater local structural variability due to the presence of defined complex filamentous boundaries.

Macro- and medium-scale contrast features (e.g., `macro_contrast_mean`, `medium_contrast_std`) also showed significant differences. These show the intensity variation patterns consistent with the morphological complexity of cording and non-cording aggregates. Together, these results demonstrate that LBP-based texture descriptors and multiscale contrast statistics can effectively capture and differentiate the phenotypic signatures of cording and non-cording aggregates.

4.5.6 Feature Correlation and Multicollinearity Analysis

Given the dominance of LBP features in the significance, it was necessary to assess the correlation of these features for inter-feature redundancy. Pairwise Spearman correlations were computed among the significant features. A total of 87 strong correlations (|r| > 0.7) were identified, revealing clear clustering among feature types (Figure 4.10).

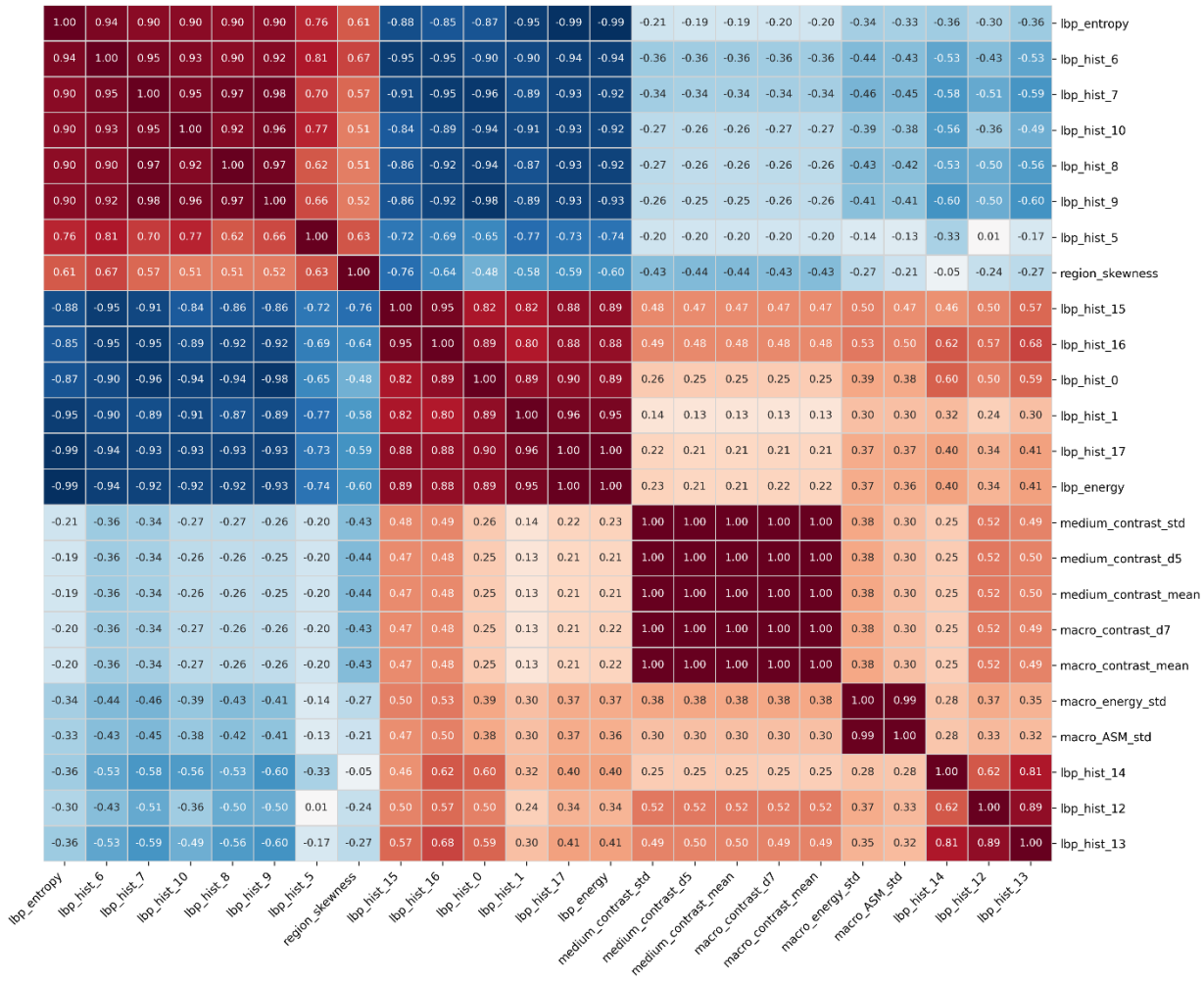


Figure 4.10: Correction Cluster Heatmap

The heatmap (Figure 4.10) visualizes correlation between the 67 texture features, with colors indicating the strength and direction of relationships. Warm colors (e.g., red) represent strong positive correlations (up to 0.999, as seen between `macro_contrast_d7` and `medium_contrast_std`), meaning these features increase together, while cool colors (e.g., blue) show strong negative correlations (down to -0.989, as between `lbp_hist_17` and `lbp_entropy`), indicating opposite trends. Values near 0 (neutral colors) suggest little to no relationship.

Multi-scale contrast descriptors (medium_contrast_d5, macro_contrast_d7, and their mean values) exhibited near-perfect correlations ($r > 0.99$), indicating that they capture highly overlapping information on image intensity heterogeneity. Within the LBP features, strong positive correlations were observed between adjacent histogram bins (e.g., `lbp_hist_7–lbp_hist_9`), reflecting co-occurring local texture motifs, while negative correlations (e.g., `lbp_hist_0–lbp_hist_9`) suggested mutually exclusive textural patterns distinguishing the two phenotypes. The LBP energy and entropy features were almost perfectly inversely correlated ($r = –0.99$), consistent with their theoretical relationship between textural uniformity and randomness. Overall, these results highlight substantial feature redundancy within specific descriptor families.

4.5.7 Feature Retention Rationale

Although several strong correlations were observed and some features showing insignificance, all features were retained for subsequent classification analysis. The primary rationale for this decision was to preserve the full spectrum of texture information captured across local (LBP histograms), regional (contrast and energy measures), and global (macro-scale and region-based) descriptors. Each feature family captures distinct yet potentially complementary aspects of phenotype morphology and reducing dimensionality at this stage could risk excluding subtle but biologically relevant texture variations.

Additionally, the subsequent classification model, SVM, is well suited for high-dimensional feature spaces. SVMs operate by finding optimal separating hyperplanes and are less affected by multicollinearity. Retaining all features therefore ensures that the model can learn non-linear relationships between correlated texture descriptors and phenotype classes, potentially enhancing generalization and discriminative performance.

4.5.8 Conclusion

The multi-scale texture analysis successfully identified distinct and statistical signatures differentiating cording (Parental_Phigh) and non-cording (Parental_Pminus) aggregates. LBP descriptors demonstrated the strongest discriminative power, revealing that cording aggregates have more homogeneous, organized textures while non-cording aggregates are characterized by having irregular granular patterns. These quantitative texture signatures provide a foundation for computational classification of mycobacterial phenotypes by offering quantitative metrics to complement traditional visual assessment of cording behavior.

4.6 Support Vector Machine Classification

Following the feature extraction and statistical analysis, a Support Vector Machine (SVM) classifier was trained to distinguish between cording and non-cording phenotypes based on the 67 texture features. The analysis utilized a dataset of 175 total samples, with 20 samples each for the labeled cording from Parental_Phigh folder and non-cording classes from Parental_Pminus folder.

4.6.1 Model Training and Optimization

The SVM model utilized all 67 extracted texture features, categorized as: 24 medium-scale GLCM, 18 macro-scale GLCM, 20 LBP, and 5 region-based statistical features. The dataset was split into **70% training (28 samples)** and **30% testing (12 samples)**, stratified to maintain balanced class representation (20 cording and 20 non-cording samples). Feature values were standardized using StandardScaler prior to model fitting.

Hyperparameter optimization was performed using `GridSearchCV` with a five-fold cross-validation scheme. The parameter grid included `C = [0.1, 1, 10, 100]`, `gamma = ['scale', 'auto', 0.1, 0.01, 0.001]`, and `kernel = ['rbf', 'linear']`. The optimal configuration was `C = 0.1`, `gamma = 'scale'`, and `kernel = 'linear'`, achieving perfect cross-validation accuracy (1.000 ± 0.000):

Table 4.4: Performance summary of SVM Training.

	Precision	**Recall**	**f1-score**	**Support**
Pminus(non-cording)	1.00	1.00	1.00	6
Phigh(cording)	1.00	1.00	1.00	6
accuracy			1.00	12
macro avg	1.00	1.00	1.00	12
weighted avg	1.00	1.00	1.00	12

The classifier achieved perfect performance on the test set (accuracy = 100%), with precision, recall, and F1-scores all equal to 1.00 for both classes. Cross-validation performance was equally stable (1.000 ± 0.000). This near-perfect classification suggests that the texture features provide strong discriminatory power, effectively capturing the phenotypic distinctions between cording and non-cording aggregates identified earlier (e.g., higher LBP energy and contrast in Pminus versus smoother, homogeneous textures in Phigh).

4.6.2 Predictions on the Full Dataset

The trained SVM was subsequently applied to 135 unseen samples that had been manually classified by direct visualization of either the samples (in the research laboratory) or by viewing the images. These manual classifications were verified by subsequent assays to discriminate between cording and non-cording cells based on a drug susceptibility phenotype. Manual classifications therefore represent the ground truth (True Label). Classification outcomes summarized in Figure 4.11. The model correctly identified 74 samples as non-cording (with 7 true non-cording samples incorrectly classified as cording) and 61 as cording (with no true cording samples classified as non-cording), achieving an overall accuracy of 94.8%.

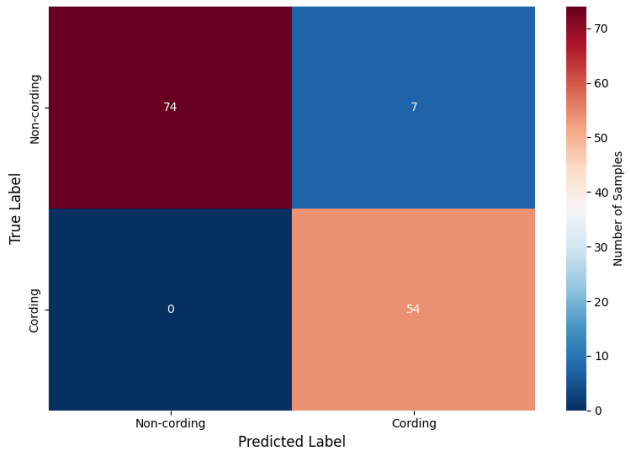


Figure 4.11: Confusion Matrix: Cording vs non-cording classification

The classifier demonstrated perfect sensitivity (recall = 1.00), correctly detecting all 54 expert-labeled cording samples with no false negatives, and high specificity (91.4%), misclassifying only 7 out of 74 non-cording samples. Precision (88.5%) and F1-score (93.9%) indicated strong predictive reliability. The average prediction confidence was 0.8, with 110 samples classified with high confidence (≥ 0.8).

The SVM classification results confirm that texture-derived quantitative features can reliably distinguish cording and non-cording phenotypes. The model's high accuracy and confidence across unseen samples underscore the discriminative strength of LBP and GLCM descriptors, providing a reproducible computational framework for cording phenotype identification.

4.6.3 Performance Using Statistically Significant Features

To assess whether dimensionality reduction based on statistical significance could improve model generalization, the SVM classifier was retrained using only the 24 texture features identified as significant through the Mann–Whitney U and FDR-adjusted analysis. The retrained model was then evaluated on the same set of 135 manually classified samples.

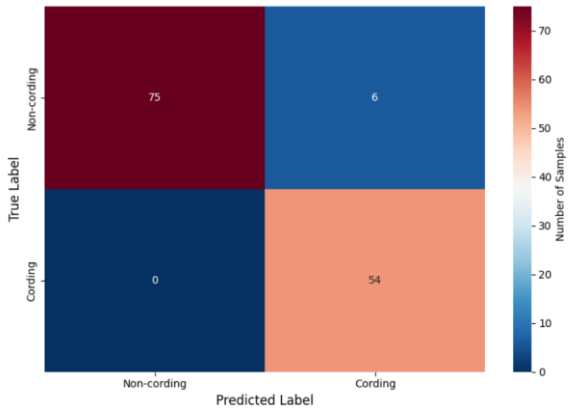


Figure 4.12: Confusion Matrix: Cording vs non-cording classification using significant features

The classifier achieved an overall accuracy of 95.6%, correctly identifying 54 cording and 75 non-cording samples, with no false negatives and six false positives. The model therefore exhibited perfect sensitivity (recall = 1.00) and high specificity (92.6%), indicating complete detection of all cording samples while maintaining low misclassification among non-cording cases. Precision (0.90) and F1-score (0.95) further reflected strong predictive reliability and class balance. Prediction probabilities were notably confident, with 116 of 135 images (85.9%) classified with probability ≥ 0.8, demonstrating model certainty in most classifications. These results show that feature selection can reduce dimensionality and preserve discriminative strength.

4.7 Chapter Summary and Conclusion

The results presented in this chapter demonstrate the successful development and implementation of a complete image processing pipeline for the automated quantification and classification of Mtb phenotypes into cording and non-cording categories. The pipeline effectively curated a dataset of 183 microscopy images, with pre-processing reducing image noise by an average of 12.4% using median filtering while preserving phenotypic structures critical for downstream analysis. The Parental_Phigh and Parental_Pminus image sets were identified as the labeled datasets representing cording and non-cording phenotypes, respectively. From these, 48 images (24 per class) were partitioned into 288 tiles (144 per class) for segmentation model training.

A U-Net segmentation model was trained on the 288 tiles with Edge-Dilated Weak Masking and achieved an acceptable performance with Dice coefficients of 0.616 and 0.614 for Phigh (cording) and Pminus (non-cording) phenotypes respectively and demonstrated good generalization with minimal overfitting.

Subsequent multi-scale Gray Level Co-occurrence Matrix (GLCM) texture analysis generated 67 quantitative features capturing medium- and macro-scale spatial dependencies, supplemented with LBP and regional descriptors. Statistical analysis using the Mann–Whitney U test with Benjamini–Hochberg FDR correction revealed distinct phenotypic texture signatures. Cording aggregates exhibited smoother, more homogeneous textures characterized by higher energy and homogeneity, while non-cording aggregates displayed greater textural variability and contrast, reflecting their granular morphology. Several features showed highly significant differences (adjusted $p < 0.000005$) and large effect sizes ($|r| \geq 0.915$), confirming their strong discriminative power.

An SVM classifier trained on all 67 features achieved perfect accuracy (100%) on the initial test set (28 samples) and 94.8% accuracy on 135 unseen samples. The classifier demonstrated perfect sensitivity (recall = 1.00) and high specificity (91.4%), with an average prediction confidence of 0.8. These results validate the model's ability to generalize well beyond the training data and highlight the effectiveness of texture-based features for phenotype discrimination. When the classifier was retrained using only the 24 statistically significant features identified through FDR correction, performance remained high (accuracy = 95.6%, recall = 1.00, specificity = 92.6%) with a comparable mean confidence level 0.8.

Overall, the results confirm that the proposed pipeline can automatically and accurately characterize Mtb cording phenotypes from non-cording based on texture-derived features. Chapter 5 will discuss the implications of these findings, evaluate the biological and statistical significance of the texture signatures, assess the effect of feature redundancy on classification performance, and investigate the SVM's bias towards cording phenotype.

Chapter 5

Discussions

5.1 Overview

This chapter discusses the results obtained from the automated image processing pipeline developed to quantify and classify Mycobacterium tuberculosis (Mtb) phenotypes. The primary objective of this project was to create a pipeline integrating data curation, image pre-processing, U-Net segmentation, feature extraction using Gray Level Co-occurrence Matrix (GLCM), and Support Vector Machine (SVM) classification, supported by statistical analysis to identify discriminative features.

This discussion aims to interpret the findings from Chapter 4 in the context of their biological and statistical significance, compare them with existing literature, and evaluate the pipeline's strengths and limitations. It will explore the implications of texture signatures for understanding Mtb phenotypes, assess the effectiveness of the statistical and machine learning approaches given high feature collinearity, and suggest ways to improve model generalization. The chapter will also investigate potential challenges, such as the small training sample size and data quality, and propose future directions to improve the pipeline.

5.2 Interpretation of Results

5.2.1 Segmentation with Weak Supervision

The U-Net model demonstrated strong segmentation performance, achieving Dice Coefficients (DC) of 0.616 and 0.614 for cording and non-cording phenotypes, respectively. In the context of medical image analysis, where a DC of 0.7 is considered excellent (Guindon & Zhang, 2016), these results are notably good, especially given the challenges of noisy input labels and limited dataset. The model's stable convergence suggests that with additional training epochs (Figure 4.4), even higher performance could be realized.

A key factor in this success was the Edge-Dilated Weak Masking (EDWM) technique. By generating coarse masks via Canny edge detection and morphological dilation, EDWM effectively overcame the issues of uneven illumination and boundary artifacts that plagued traditional methods like Otsu thresholding. This approach enabled stable training, with validation loss plateauing between 0.150-0.200 after 50 epochs and IoU metrics reaching 0.575 (cording) and 0.574 (non-cording). The slight discrepancy between class-specific and overall scores suggests that the current model may have limited generalization to samples, where both cording and non-cording features co-occur. Expanding the training dataset to include such ambiguous cases could improve overall performance.

The success of weak mask supervision is very useful in Mtb phenotype analysis. This is because creating precise manual annotations is labour-intensive and prone to subjectivity. The model's ability to produce refined masks with smooth boundaries and minimal cross-prediction (zero overlap in class probabilities, as seen in Figure 4.5) demonstrates that EDWM captures essential morphological structures. Qualitatively, minor under-segmentation observed in dense regions (Figure 4.6(b)) show some challenges in weak labels, however, the segmented regions preserved phenotypic integrity for downstream texture extraction. Under-segmented non-cording regions may have appeared texturally richer due to these artifacts, amplifying LBP feature responses relative to the more sharply delineated cording structures. This suggests that segmentation quality, particularly at region boundaries, may influence the apparent discriminative power of texture-based features. Compared to fully supervised models requiring pixel-perfect labels, this method reduced annotation effort while maintaining comparable Dice scores to literature

benchmarks which is 0.6-0.7 in biological studies (Guindon & Zhang, 2016). These findings align with the validation of Hypothesis H2, which stated that weakly-supervised segmentation would achieve Dice coefficients greater than 0.60 while preserving biological relevance, as assessed in this section.

5.2.2 Biological Relevance of Texture Signatures

The multi-scale texture profiles reveal biologically meaningful distinctions between Mtb phenotypes. Cording's lower medium-scale contrast (19.1758 ± 11.2385) and higher homogeneity (0.9361 ± 0.0468) suggest organized, branched, rope-like structures, which can be potentially linked to virulence and drug susceptibility in Mtb (Julián, et al., 2010). Conversely, non-cording's higher macro-scale contrast (45.2361 ± 22.6855) and LBP energy (0.1562 ± 0.0080) indicate irregular, scattered aggregates with greater intensity variations, which are consistent with disrupted, fragmented structures. These patterns, dominated by LBP histogram bins (middle bins 7-9 for cording edges, higher bins 15-17 for non-cording complexity), may reflect microstructural differences which could provide a quantitative insight for phenotype assessment beyond simple visual observation on a microscope.

5.2.3 Statistical and Effect Size Considerations

The most discriminative texture features exhibited consistently high effect sizes ($|r| \geq 0.915$) and remained statistically significant following FDR correction (adj. $p < 0.000005$), indicating that the observed differences represent substantial, biologically meaningful separations between cording and non-cording phenotypes rather than minor statistical artefacts. The use of rank-biserial correlation as an effect size measure provided an appropriate metric for non-parametric data, capturing both the strength and direction of group separation.

Initially, all 67 extracted features were retained for classification despite detectable collinearity, justified by the classifier's strong generalization performance (accuracy = 94.8%) and high confidence levels across unseen data. However, subsequent analysis demonstrated that a reduced subset of 24 statistically significant features achieved comparable performance (accuracy = 95.6%), confirming that the discriminative information was largely concentrated within this refined feature set. This finding reinforces that, while some redundancy exists among texture descriptors, their collective representation provides a biologically interpretable signature of cording behavior, supporting Hypothesis H1, which stated that multi-scale texture features demonstrate statistically significant differences between cording and non-cording phenotypes.

5.2.4 SVM Performance Analysis

An SVM classifier trained on all 67 extracted features achieved 94.8% accuracy and perfect recall (100%) across 135 unseen samples. The model correctly identified all 61 cording samples, with no false negatives, and misclassified seven non-cording samples as cording, yielding a specificity of 91.4%. The average prediction confidence was 0.80, with 110 samples (81.5%) showing high-confidence classifications (probability ≥ 0.8). These results demonstrate that texture-derived features can distinguish cording phenotypes and confirm the model's strong generalization to unseen data.

To assess whether model simplification could enhance interpretability and stability, a second SVM classifier was trained using the 24 most statistically significant texture features ($|r| \geq 0.915$, adj_$p < 0.000005$). This reduced-feature model achieved slightly higher performance, with an overall accuracy of 95.6% (129/135), precision of 0.90, recall of 1.00, F1-score of 0.947, and specificity of 92.6%. Importantly, 116 samples (85.9%) were classified with high confidence (probability ≥ 0.8), indicating greater certainty in predictions relative to the full-feature model. These findings demonstrate that the most discriminative multi-scale texture features, derived from LBP histograms and GLCM descriptors, were sufficient to capture the defining characteristics of the cording phenotype.

The complete absence of false negatives in both models is particularly significant within the context of mycobacterial diagnostics, where undetected cording phenotypes could lead to underestimation of virulence or misinterpretation of drug susceptibility. The small number of false positives suggests a conservative classification bias toward detecting cording-like structures. This tendency likely arises from three interacting factors:

1. Biological Heterogeneity: Non-cording populations may exhibit phenotype variations that may share some texture characteristics with cording phenotypes. The false positives could represent non-cording samples with partial organization or aggregation patterns that the model interprets as cording signatures.

2. Training Data Characteristics: The use of control experiment data cording (Parental_Phigh) and non-cording (Parental_Pminus) strains for training may have created a model optimized for clear cases and may potentially struggle with borderline cases.

3. Feature Sensitivity: LBP histograms were the most discriminative features for classification. These features may be more sensitive to cording patterns than to non-cording, which could create a bias towards cording detection.

Both models performed strongly, but the reduced 24-feature model provided improved accuracy, higher classification confidence, and greater interpretability without loss of sensitivity. The findings therefore support Hypothesis H3, confirming that the integrated pipeline generalizes effectively to unseen data with a classification accuracy exceeding 90%. The optimized model prioritizes biological interpretability by using statistically significant features only, offering a practical and reproducible computational framework for Mtb phenotype classification.

5.3 Comparison with existing literature

5.3.1 Segmentation Techniques

The U-Net segmentation model achieved a Dice coefficient 0.616 for cording and 0.614 for non-cording phenotypes, and corresponding Intersection-over-Union (IoU) values of 0.575 and 0.574, respectively. Traditional methods like Otsu thresholding struggled with uneven lighting and boundary definition in the dataset. In contrast, (Núñez-Fernández, et al., 2020) reported a more advanced U-Net-based system for automatic segmentation of tuberculosis cords using lens-free microscopy images, which achieved an IoU of 0.88.

The presented model's IoU is notably lower than the 0.88 reported by Núñez-Fernández et al. (2020) due to differences in imaging modalities and annotation quality. The use of Edge-Dilated Weak Masking (EDWM) in the pipeline, which generated coarse masks enabled effective training with minimal annotation effort, which could be the main difference with the potentially more refined labels used in the referenced work. The slight under-segmentation observed and lower IoU indicate that the model may benefit from enhanced label quality or additional training data.

5.3.2 Texture Analysis

The multi-scale GLCM-LBP approach which generated 67 texture features, provided a detailed characterization of Mtb phenotypes. In prior work, Hlabangana (2024) employed image processing techniques for pre-processing to extract shape measurements and texture features, with a primary focus on Haralick features, to support a binary classifier for phenotype classification and achieved a 93% accuracy on test data which demonstrated the effectiveness of texture-based approaches in Mtb phenotyping. While Haralick features, typically consisting of 13-14 descriptors, have been widely applied in medical imaging, the proposed GLCM incorporates multiple-spatial scales and 20 LBP features, which result in a more comprehensive 67-feature set. The multi-spatial scale captured fine structural patterns and broader organizational features, which offered greater granularity.

The distinct texture signatures identified; smoother homogeneity in cording versus higher contrast and LBP energy in non-cording, build on the texture-phenotype correlations established in prior work (Hlabangana 2024). The novelty of LBP histogram insights, with middle bins (e.g., lbp_hist_7-9) dominating cording and higher bins (e.g., lbp_hist_15-17) in non-cording, adds a microstructural perspective that was not emphasized in Hlabangana's Haralick-centric study.

Performance evaluation demonstrated that the SVM trained on the full 67-feature set achieved 94.8% accuracy, while a model using only the 24 most significant texture features achieved 95.6% accuracy, both outperforming the earlier

benchmark of 93% reported by Hlabangana (2024). The improvement likely reflects the greater feature diversity and spatial scale integration, which enhance the model's ability to capture biologically meaningful variations in texture. Hlabangana's inclusion of morphological shape features, however, suggests that a hybrid model combining both shape and multi-scale texture information could potentially improve classification, particularly for phenotypes with intermediate or ambiguous morphologies.

The results establish that extending Haralick-based texture analysis to a multi-scale GLCM–LBP framework improved classification accuracy and the biological interpretability of texture–phenotype relationships in Mtb.

5.4 Limitations

5.4.1 Sample Size and Generalizability

One of the main limitations of this study was the relatively small training dataset, consisting of only 28 samples (20 per class for Parental_Phigh and Parental_Pminus) used to train the SVM classifier, despite a total dataset of 175 samples. This small sample size limits the model's ability to capture the full variability of phenotypes, potentially leading to overfitting. The 94,8% accuracy on unseen data shows promise, however the reliance on a limited training set may not fully represent the diversity encountered in a research setting. Retraining with 24 statistically significant features slightly improved accuracy to 95.6%, suggesting that focused selection can mitigate some effects of limited data, but larger datasets remain essential for better generalization.

5.4.2 Feature Collinearity

The high collinearity among the 67 texture features, with correlations exceeding 0.9 introduces a challenge. While keeping all features preserved the complete texture spectrum, the redundancy overemphasizes correlated variables which could potentially mask the true discriminative power of the individual features. Collinearity can also affect the statistical analysis by skewing effect sizes. It might therefore be necessary to explore data dimensionality reduction, especially given the small training dataset. The retrained SVM using 24 significant features achieved 95.6% accuracy, indicating that collinearity can be addressed through targeted selection, though this highlights the need for techniques like PCA to optimize the feature space further.

5.4.3 Data Quality and Labelling

The use of Edge-Dilated Weak Masking (EDWM) for U-Net segmentation, while effective, introduces potential data quality issues due to the noisy nature of weak labels. Although EDWM outperformed Otsu thresholding, the slight under-segmentation minor false positives/negatives suggest that the inaccurate labels may have affected segmentation accuracy. They may have also been worsened by using only 48 images (24 per phenotype) for training the segmentation model, which may not fully capture the variability across the entire dataset. Additionally, the pre-processing step reduced noise by 12.4% through median filtering, but some artifacts remained in raw images (RMS contrast 6.67-8.49) which could still affect downstream feature extraction. There is therefore a need for more diverse, high-quality labeled training data and accurately annotated masks to enhance model performance.

Chapter 6

Conclusion

This study has successfully developed and evaluated an automated image processing pipeline for the quantitative analysis of Mtb cording phenotypes, addressing a significant gap in microbiological diagnostics and research. By leveraging a novel dataset of 183 low-magnification images, the pipeline integrates U-Net-based segmentation with EDWM, multi-scale texture feature extraction using GLCM and LBP methods, and SVM classification. The segmentation model achieved high class-specific Dice coefficients (0.887 for Parental_Phigh [cording] and 0.851 for Parental_Pminus [non-cording]), reflecting robust performance in phenotype-specific contexts, although the overall Dice (0.616/0.614) highlighted challenges in multi-class generalization due to class imbalance.

Texture analysis identified 24 significant features out of 67, with LBP histograms proving most discriminative, supporting the SVM's 94.8% accuracy on 135 unseen samples, perfect recall (1.00), and high confidence (average 0.8, 81.5% $\geq$ 0.8). Retraining with these 24 features improved accuracy to 95.6%, demonstrating enhanced interpretability without loss of sensitivity. The experimental design, incorporating FDR-corrected statistical tests, provided a rigorous foundation, revealing the biological relevance of texture signatures—smoother homogeneity in cording vs. higher contrast in non-cording—that may link to virulence and drug susceptibility.

Despite these successes, limitations such as the small training set (48 labeled images) and reliance on weak masks suggest areas for improvement. Future work should focus on expanding the dataset with mixed-phenotype images, refining segmentation through enhanced masks, and exploring advanced classifiers and feature reduction to mitigate overfitting risks. This study lays groundwork for automated Mtb phenotyping, with potential for clinical diagnostic tools and contributions to ongoing research on Mtb mechanopathology.

Chapter 7

Further Work

7.1 Enhancing Model Generalizability

To address the limitations of the small training dataset and potential overfitting, future efforts should expand sample size by adding labelled images or use synthetic data augmentation techniques. Increasing the training set could better capture the phenotypic variability of Mtb phenotypes, enhancing the U-net and SVM classifier's generalizability. Synthetic data generation, such as applying geometric transformations or noise perturbations to existing images, could supplement the 183-image dataset. Additionally, applying dimensionality reduction techniques such as Principal Component Analysis (PCA) to mitigate the high collinearity among the 67 texture features could refine feature selection, reduce redundancy and potentially decrease the 7 false positives observed, thereby improving the model's precision for research applications. It is also critical to train both the U-net and SVM using a more diverse dataset with more ambiguous cases of cording and non-cording phenotypes.

7.2 Advanced Classification Techniques

Alternative classifiers beyond the current SVM could be explored to validate and potentially enhance phenotype classification performance. Gradient Boosting models, known for handling high-dimensional data effectively, or Deep Learning approaches like Convolutional Neural Networks, could be tested to leverage the rich texture information. Ensemble methods, combining SVM with other algorithms (e.g., Random Forests), might reduce the slight bias toward cording predictions (evidenced by perfect recall but 91.4% specificity) and improve overall specificity. These advancements could be evaluated on an expanded dataset, with hyperparameter tuning tailored to the 67-feature space or the reduced 24-feature set (which improved accuracy to 95.6%), offering a pathway to surpass the current 94.8% accuracy and address overfitting concerns.

Appendix A

A1. GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year report can be found.

GitHub repository	https://github.com/vinny-mze/MycoMorph.git

Appendix B

B1. Ethics approval



PRE-SCREENING QUESTIONNAIRE OUTCOME LETTER

STU-EBE-2025-PSQ001909

2025/08/27

Dear Vincent Muzerengwa,

Your Ethics pre-screening questionnaire (PSQ) has been evaluated by your departmental ethics representative. Based on the information supplied in your PSQ, it has been determined that you do not need to make a full ethics application for the research project in question.

You may proceed with your research project titled:

Automated quantitative analysis of mycobacterial cording phenotypes

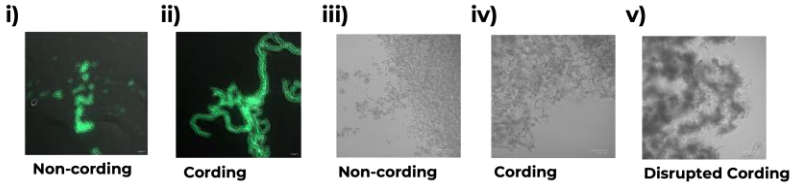
Please note that should aspect(s) of your current project change, you should submit a new PSQ in order to determine whether the changed aspects increase the ethical risks of your project. It may be the case that project changes could require a full ethics application and review process.

Regards,

Faculty Research Ethics Committee

Appendix C

C1. Topic description: EEE4022S RAV2025-06

Student Proposed?	**N**	*N/A*
ID	**RAV2025-06**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr MK Mason (MMRU, IDM)**	
Title	**Automated quantitative analysis of mycobacterial cording phenotypes**	
Description	Quantitative data acquisition and analysis is a key component of microbiological sciences, requiring critical technological skills in sample processing, data generation, acquisition, digitisation, processing and statistical analysis. *Mycobacterium tuberculosis*, the bacterium that is responsible for causing tuberculosis, has the ability to form cords when cultured in the laboratory under specific growth conditions. Cording is clinically relevant, and has been shown to contribute to the mechanopathology of *Mtb* (**Mishra** et al., 2023). Cords have been characterised at high resolution using scanning electron microscopy (**Julián** et al., 2010) but are also visible microscopically at lower resolutions, can be used as a diagnostic (**Caviedes** et al., 2000) and are a phenotype of interest (see Figure 6).  **Fig. 6.** (Microscopy images of *Mycobacterium tuberculosis* aggregations to illustrate cording and non-cording phenotypes. High resolution of **(i)** a non-cording strain showing a disorganised bacterial aggregate and **(ii)** a cording strain showing an organised stranded structure (Green indicates fluorescence due to DMNTre labelling of mycolic acids). Lower magnification grey scale images show **(iii)** non-cording **(iv)** and cording phenotypes with **(v)** the disruption of cording shown in a cording strain that has lost a critical cell envelope component. (Image credit: Dr MK Mason, 2024) Microscopic cording has generally been assessed qualitatively through the comparison of high-resolution images or through the visual categorisation of phenotypes into cording and non-cording (e.g. for the diagnostic MODs assay) (**Caviedes** et al., 2000). For this project, low magnification images of mycobacterial cells in non-coding aggregates (see Fig. 6, **iii**), organised cords (see Fig. 6, **iv**), and disrupted cords (see Fig. 6, **v**) have been generated as phenotypic readouts of microbiological assays (background strain phenotypes and disruption of cell envelope process). These data form the foundation for a qualitative description of the multicellular organisation of these specific strains and conditions. The aim of this project would be to assess these images and to design and develop an automated image processing, data acquisition, analysis and visualisation solution, which would allow for the quantification and comparison of data for each strain and condition. If more suited, a machine learning approach could be implemented for the classification of distinct phenotypes into specific categories.	

Deliverables	**The key objectives of this project are:** (a) to understand the full scope and requirements of the project. (b) to conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. (c) to familiarise self with image dataset and curate dataset in a manner appropriate to automated data processing (file re-organisation, re-naming, categorisation/labelling). (d) to explore image analysis techniques that allow for the quantification of complex features and textures. (e) to pre-process image data to allow for uniformity background (to accommodate acquisition artifacts and allow standardisation within and among experiments) (f) to process image data to allow bacterial aggregate or cord feature recognition and data acquisition. (g) to quantify cording phenotypes / features to allow for analysis (comparisons among samples) (h) OPTIONAL: explore and implement a machine learning approach to classify images in terms of their cording phenotype. (i) test and evaluate the system performance and compare to previously implemented solutions. (j) discuss the performance of the system, draw conclusions and make recommendations for future improvements. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project) (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed upon communication platform, which could include online meetings (c) An image pre-processing, processing and data analysis pipeline that is suited for the quantification of cording phenotypes for the images provided. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs) (f) A poster summarising the research project report (g) A seminar to the Department summarising the research project report (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Python, Mathematical skills, Data analysis and visualisation, Statistical competencies, Interest in biomedical engineering
Extra Information	**References** L. **Caviedes**, T. S. Lee, R. H. Gilman, P. Sheen, E. Spellman, E. H. Lee, D. E. Berg, and S. Montenegro-James. Rapid, efficient detection and drug susceptibility testing of *Mycobacterium tuberculosis* in sputum by microscopic observation of broth cultures. *Journal of Clinical Microbiology*, 38(3):1203–1208, Mar 2000. **doi:** 10.1128/JCM.38.3.1203-1208.2000 E. **Julián**, M. Roldán, A. Sánchez-Chardi, O. Astola, G. Agustí, and M. Luquin. Microscopic cords, a virulence-related characteristic of *Mycobacterium tuberculosis*, are also present in nonpathogenic mycobacteria. *Journal of Bacteriology*, 192(7):1751–1760, Apr 2010. **doi:** 10.1128/JB.01485-09 R. **Mishra**, M. Hannebelle, V. P. Patil, A. Dubois, C. Garcia-Mouton, G. M. Kirsch, M. Jan, K. Sharma, N. Guex, J. SordetDessimoz, J. Perez-Gil, M. Prakash, G. W. Knott, N. Dhar, J. D. McKinney, and V. V. Thacker. Mechanopathology of biofilm-like *Mycobacterium tuberculosis* cords. *Cell*, 186(23):5135–5150.e28, Nov 2023. **doi:** 10.1016/j.cell.2023.09.016 **Potential for Publication** The dataset provided by Dr Mason comprises a set of low magnification images of *Mycobacterium tuberculosis* CRISPRi hypomorph strains that have been grown in liquid culture. These images represent a novel dataset of bacterial strains that are being characterised for publication. The demonstration of a tractable and complete implementation (a full, user friendly working solution) to automatically quantify (or categorise) phenotypes has the potential to complement (and contribute to) data that are planned for publication. Work performed in this project that is suited for inclusion in this manuscript will result in co-authorship for this publication. Students who are interested in contributing to publishable research and who would be interested in being involved in the publication process are encouraged to apply for this project.
Broad Research Area	Image processing and analysis, Machine Learning, Data analysis and visualisation, Microbiology, Tuberculosis.
Project suitable for: ME/ECE/EE/ALL?	ECE

Back to Topic List

EEE4022S Topic RAV2025-06 GAs

Student Proposed?	**N**	*N/A*
ID	**RAV2025-06**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr MK Mason (MMRU, IDM)**	
Title	**Automated quantitative analysis of mycobacterial cording phenotypes**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	The images provided represent a complex problem for preprocessing as they: • comprise complex imaging data. • are not standardised (contain large variations and artefacts). The data generated present a complex problem for processing as they: • Reflect data of a biological origin that may be unfamiliar and require the candidate to learn about a new field (new concepts). • Must be examined in several ways to reach the best solution for how to segment highly variable features. The data analysis component will also be complex and require novel solutions as: • The quantification of cords is an open-ended problem. • Statistical analysis of data is desirable. • Image data that have been generated are highly complex and variable and represent a novel phenotype and various analysis approaches must be explored.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Explore segmentation, feature identification approaches to determine if/how data can be quantified. • Explore machine learning approaches to determine, if these should be utilised, to categorise data (strains and conditions) into meaningful groups.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This topic, which provides an image dataset for processing and analysis, will be computational in nature and will require competencies in file handling, image processing, data handling, and data visualisation. Due to the complex nature of the images provided, it is suggested that a Machine Learning approach could be explored for a categorisation approach. As such, the candidate will demonstrate competency in the use of computing and software and must select the appropriate data handling processes and software tools to accomplish their objectives. **Software required may include:** • Python/MATLAB • ImageJ/FIJI	

*

Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues; and their solutions have one or more of the characteristics:
  - are not obvious, require originality or analysis based on fundamentals;
  - are outside the scope of standards and codes;
  - require information from a variety of sources that is complex, abstract or incomplete;
  - involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-06

Question	Question Text	Yes	No
Q1	Does this project involve data collection		**X**
Q2	Does this project involve utilizing a third-party data set	**X**	
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?	**X**	
Q4	Does it exceed the minimum risk defined **here**?		**X**
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		**X**
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		**X**
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)	**X**	
Q8	Are the third-party data used from an open source?		**X**
Q9	Are the third-party data used from a different research group?		**X**
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		**X**
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		**X**
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		**X**

Appendix D: ECSA Graduate Attributes

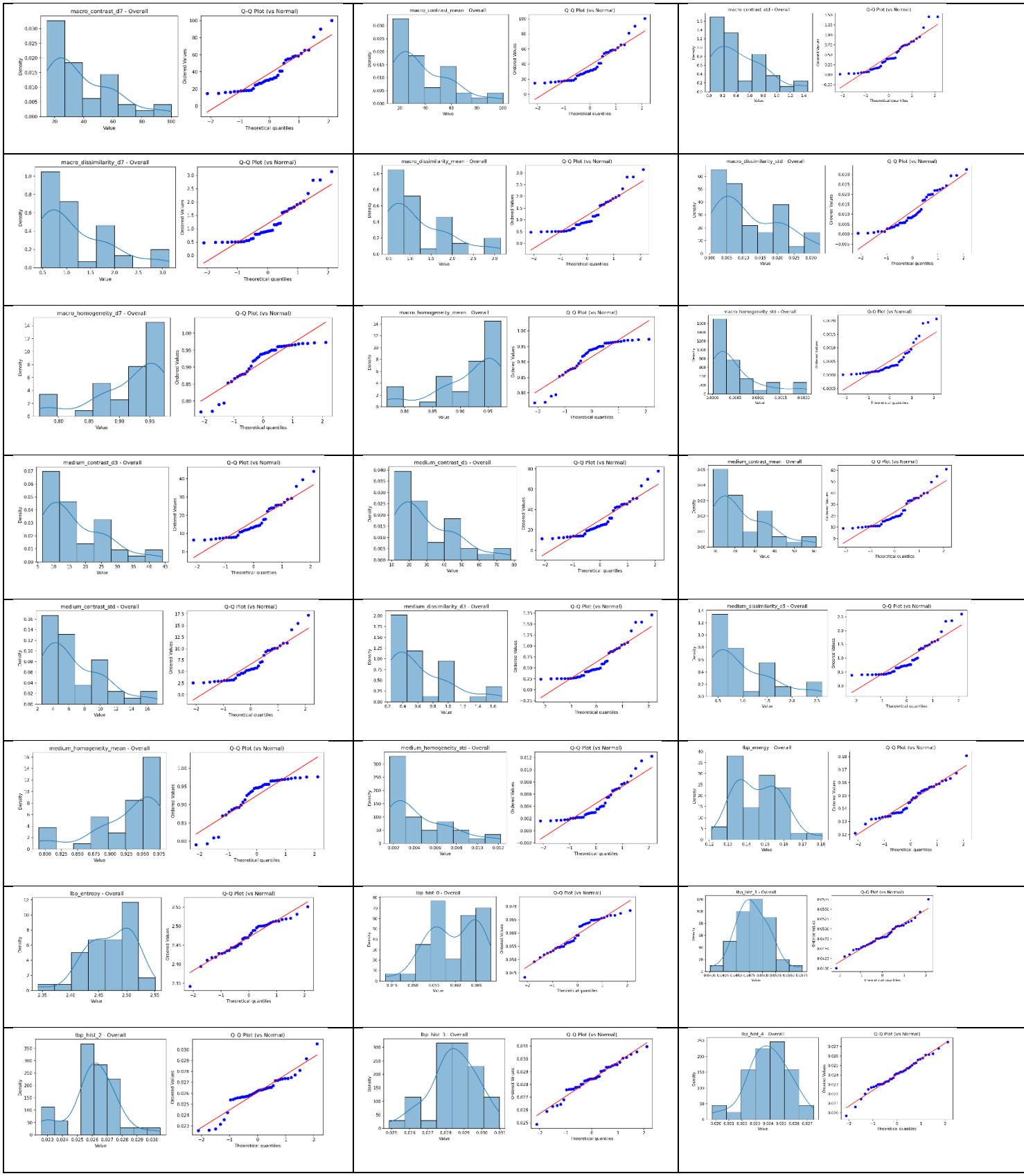
A.1 Project mapping to Engineering Council of South Africa (ECSA) Graduate Attributes (GAs)

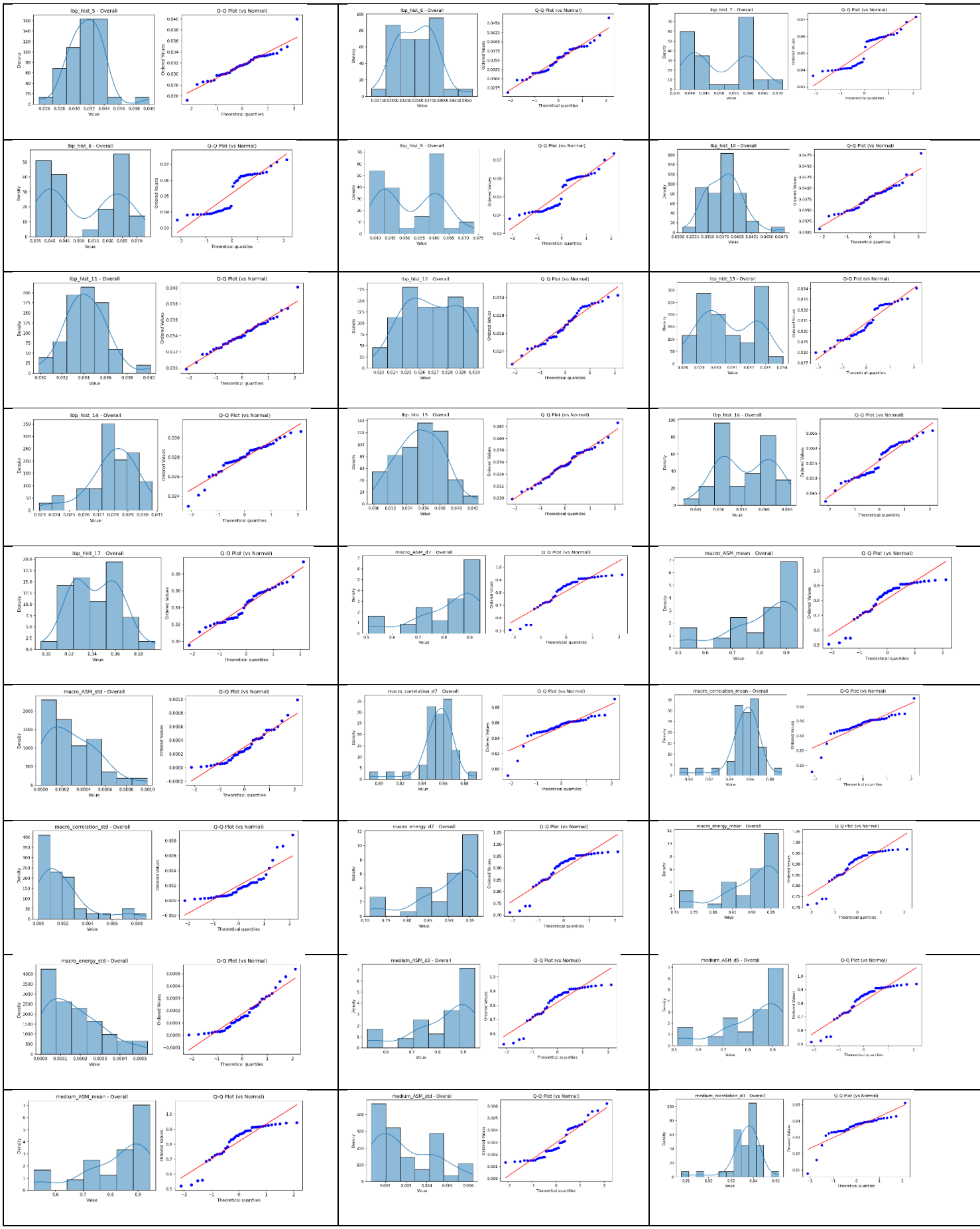
The table below provides a structured reflection on where each of the required GAs as defined by ECSA associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

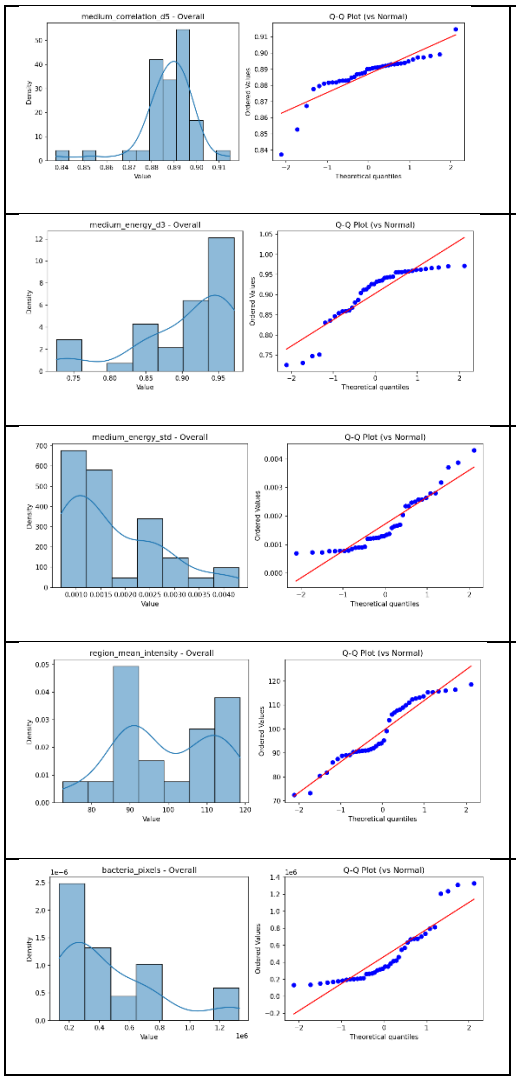
GA	GA description	Student reflection: How was this outcome achieved during the project
GA1	*Problem solving*	I demonstrated problem-solving skills throughout the project. Initially, the absence of annotated segmentation masks posed a major limitation to training a supervised model. This was overcome through the design and implementation of a weakly supervised "Edge Dilated Weak Masking (EDWM)" method, which generated approximate ground truth masks by combining edge detection and morphological dilation — an innovative solution. In the analysis phase, on of the challenge I faced was non-normality distribution of features. To address this, I used non-parametric testing and False Discovery Rate (FDR) correction which ensured that my method was statistically robust. I also managed to integrate a segmentation mode, a feature extractor and a classifier model into a unified system with minimum user input. This demonstrated my ability to design a complex, multi-stage computational workflow.
GA4	*Investigations, experiments and data analysis*	Investigations: I began the project with a thorough investigation. In my literature review (Chapter 2), I at manual phenotyping, and I identified its limitations, such as subjectivity between inter-observers which informed me of the need for a quantitative texture-based approach. Experiments: I performed experiments in the segmentation phase where I compared otsu-thresholding with EDWM to address the lack of well-delineated label masks. Comparative evaluations demonstrated improved mask consistency. Analysis: I extracted texture features using multi-scale GLCM and LBP descriptors. I then conducted statistical analyses, including normality testing (Shapiro–Wilk), non-parametric hypothesis testing (Mann–Whitney U), and multiple testing correction (Benjamini–Hochberg FDR), to identify significant discriminators between phenotypes. (Chapter 3.6)
GA5	*Use of engineering tools*	Python (Tensorflow, OpenCV, NumPy, scikit-learn) Google Colaboratory Image processing techniques (Chapter 3)
GA6	*Professional and technical communication*	I wrote and compiled this report following the recommended final year project report guidelines. Submission of this report is evidence that I have met this GA
GA8	*Individual and collaborative teamwork*	This final year project is an individual effort, other than the normal guidance of my supervisor(s). All contributions and assistance from other parties/applications are acknowledged in my report acknowledgements and cited, where appropriate.
GA9	*Independent learning ability*	Skills gained: 4. I learned how to use Applied statistics for analysis in bacterial phenotyping. 5. An improved understanding of machine learning algorithms, convolutional neural networks specifically. 6. An understanding of Mtb and diagnostic methods used for detection of Mtb. 7. A deeper understanding of image processing techniques

Appendix E

E1. Feature Distribution Plots




lbp_hist_5 - Overall
Q-Q Plot (vs Normal)
lbp_hist_6 - Overall
Q-Q Plot (vs Normal)
lbp_hist_7 - Overall
Q-Q Plot (vs Normal)
lbp_hist_8 - Overall
Q-Q Plot (vs Normal)
lbp_hist_9 - Overall
Q-Q Plot (vs Normal)
lbp_hist_10 - Overall
Q-Q Plot (vs Normal)
lbp_hist_11 - Overall
Q-Q Plot (vs Normal)
lbp_hist_12 - Overall
Q-Q Plot (vs Normal)
lbp_hist_13 - Overall
Q-Q Plot (vs Normal)
lbp_hist_14 - Overall
Q-Q Plot (vs Normal)
lbp_hist_15 - Overall
Q-Q Plot (vs Normal)
lbp_hist_16 - Overall
Q-Q Plot (vs Normal)
lbp_hist_17 - Overall
Q-Q Plot (vs Normal)
macro_ASM_d7 - Overall
Q-Q Plot (vs Normal)
macro_ASM_mean - Overall
Q-Q Plot (vs Normal)
macro_ASM_std - Overall
Q-Q Plot (vs Normal)
macro_correlation_d7 - Overall
Q-Q Plot (vs Normal)
macro_correlation_mean - Overall
Q-Q Plot (vs Normal)
macro_correlation_std - Overall
Q-Q Plot (vs Normal)
macro_energy_d7 - Overall
Q-Q Plot (vs Normal)
macro_energy_mean - Overall
Q-Q Plot (vs Normal)
macro_energy_std - Overall
Q-Q Plot (vs Normal)
medium_ASM_d3 - Overall
Q-Q Plot (vs Normal)
medium_ASM_d5 - Overall
Q-Q Plot (vs Normal)
medium_ASM_mean - Overall
Q-Q Plot (vs Normal)
medium_ASM_std - Overall
Q-Q Plot (vs Normal)
medium_correlation_d1 - Overall
Q-Q Plot (vs Normal)



E2. Table of all features and their significance

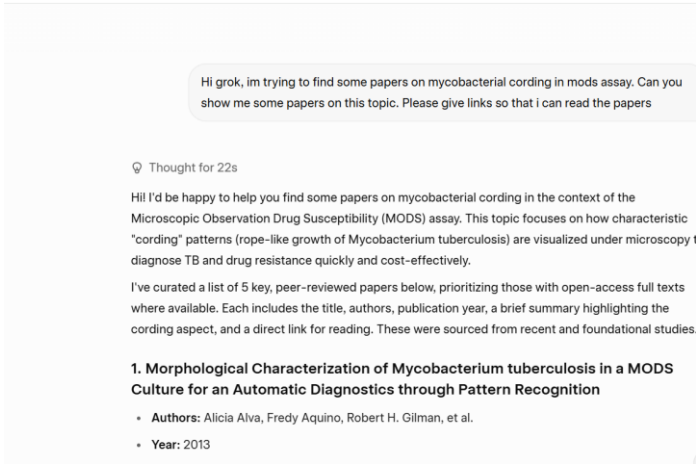
Feature	Phigh_Mean	Phigh_SD	Pminus_Mean	Pminus_SD	P_Value	P_Value_FDR	Effect_Size	Significant_FDR
lbp_hist_8	0.0637	0.0039	0.0400	0.0020	0.000000	0.000000	7.6502	True
lbp_hist_7	0.0603	0.0039	0.0417	0.0023	0.000000	0.000000	5.8233	True
lbp_hist_9	0.0613	0.0042	0.0431	0.0023	0.000000	0.000000	5.4026	True
lbp_hist_16	0.0504	0.0026	0.0611	0.0024	0.000000	0.000000	-4.3281	True
lbp_hist_0	0.0542	0.0034	0.0647	0.0020	0.000000	0.000000	-3.6987	True
lbp_hist_17	0.3257	0.0120	0.3602	0.0117	0.000000	0.000000	-2.9229	True
lbp_energy	0.1361	0.0066	0.1562	0.0080	0.000000	0.000000	-2.7316	True
lbp_hist_6	0.0383	0.0026	0.0321	0.0022	0.000000	0.000000	2.5882	True
lbp_hist_15	0.0335	0.0017	0.0377	0.0017	0.000000	0.000000	-2.4764	True
lbp_hist_13	0.0295	0.0008	0.0320	0.0012	0.000000	0.000000	-2.4139	True
lbp_hist_12	0.0253	0.0011	0.0284	0.0014	0.000000	0.000000	-2.4134	True

Feature	Phigh_Mean	Phigh_SD	Pminus_Mean	Pminus_SD	P_Value	P_Value_FDR	Effect_Size	Significant_FDR
lbp_entropy	2.5037	0.0235	2.4384	0.0315	0.000000	0.000000	2.3504	True
lbp_hist_10	0.0400	0.0025	0.0357	0.0020	0.000001	0.000004	1.9077	True
lbp_hist_1	0.0460	0.0022	0.0504	0.0026	0.000002	0.000009	-1.8265	True
lbp_hist_14	0.0272	0.0014	0.0287	0.0015	0.002298	0.010417	-1.0605	True
macro_ASM_std	0.0002	0.0002	0.0004	0.0002	0.004545	0.019315	-0.9786	True
region_skewness	-0.4236	0.1860	-0.6154	0.2103	0.005023	0.020092	0.9664	True
macro_energy_std	0.0001	0.0001	0.0002	0.0001	0.006291	0.023767	-0.9385	True
lbp_hist_5	0.0326	0.0023	0.0306	0.0021	0.008271	0.029602	0.9040	True
macro_homogeneity_std	0.0003	0.0003	0.0007	0.0007	0.040233	0.134925	-0.6891	False
medium_contrast_std	5.4309	3.1749	7.8370	3.8551	0.042421	0.134925	-0.6813	False
macro_contrast_d7	31.4093	18.3276	45.2361	22.6855	0.045636	0.134925	-0.6705	False
macro_contrast_mean	31.4093	18.3276	45.2361	22.6855	0.045636	0.134925	-0.6705	False
medium_contrast_d5	24.5166	14.3569	35.0777	17.6928	0.050431	0.139508	-0.6555	False
medium_contrast_mean	19.1758	11.2385	27.4145	13.8786	0.051466	0.139508	-0.6524	False
medium_contrast_d3	13.8349	8.1202	19.7514	10.0649	0.053341	0.139508	-0.6470	False
lbp_hist_11	0.0337	0.0020	0.0347	0.0017	0.086750	0.218482	-0.5705	False
medium_ASM_std	0.0026	0.0013	0.0033	0.0014	0.097737	0.236914	-0.5508	False
medium_energy_std	0.0015	0.0009	0.0020	0.0010	0.117187	0.236914	-0.5201	False
macro_dissimilarity_d7	1.0078	0.6356	1.3683	0.7491	0.118052	0.236914	-0.5189	False
macro_dissimilarity_mean	1.0078	0.6356	1.3683	0.7491	0.118052	0.236914	-0.5189	False
medium_dissimilarity_std	0.1510	0.0928	0.2011	0.1044	0.126175	0.236914	-0.5073	False
medium_dissimilarity_d3	0.5347	0.3485	0.7305	0.4200	0.126184	0.236914	-0.5073	False
medium_dissimilarity_mean	0.6836	0.4399	0.9280	0.5231	0.127385	0.236914	-0.5057	False
medium_dissimilarity_d5	0.8325	0.5313	1.1254	0.6263	0.128265	0.236914	-0.5045	False
medium_correlation_d3	0.9347	0.0087	0.9380	0.0043	0.150865	0.236914	-0.4756	False
macro_ASM_d7	0.8388	0.1104	0.7834	0.1271	0.160319	0.236914	0.4646	False
macro_ASM_mean	0.8388	0.1104	0.7834	0.1271	0.160319	0.236914	0.4646	False
medium_ASM_d5	0.8423	0.1087	0.7879	0.1254	0.161492	0.236914	0.4633	False
medium_ASM_mean	0.8448	0.1075	0.7912	0.1241	0.162376	0.236914	0.4623	False
medium_ASM_d3	0.8474	0.1063	0.7944	0.1228	0.163290	0.236914	0.4612	False

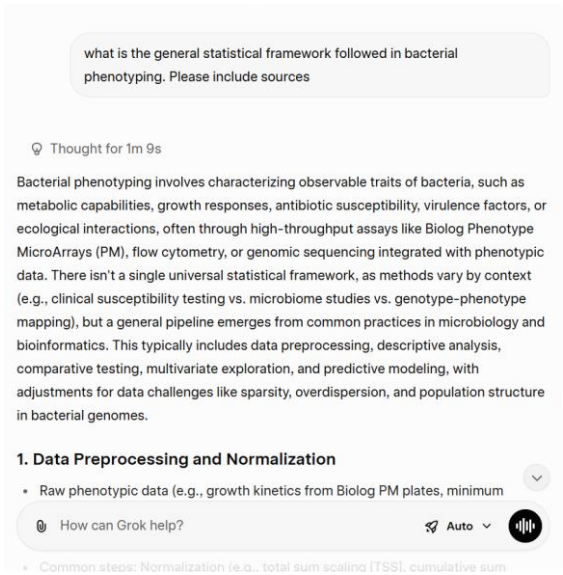
Feature	Phigh_Mean	Phigh_SD	Pminus_Mean	Pminus_SD	P_Value	P_Value_FDR	Effect_Size	Significant_FDR
macro_energy_d7	0.9136	0.0637	0.8819	0.0753	0.169318	0.236914	0.4546	False
macro_energy_mean	0.9136	0.0637	0.8819	0.0753	0.169318	0.236914	0.4546	False
medium_energy_d5	0.9156	0.0626	0.8846	0.0740	0.170373	0.236914	0.4534	False
medium_energy_mean	0.9171	0.0617	0.8865	0.0731	0.171163	0.236914	0.4525	False
medium_energy_d3	0.9185	0.0609	0.8884	0.0721	0.171981	0.236914	0.4517	False
medium_correlation_mean	0.9095	0.0120	0.9138	0.0060	0.173137	0.236914	-0.4504	False
region_area	392799.6000	296798.4602	539297.7000	352347.8244	0.173794	0.236914	-0.4497	False
bacteria_pixels	392799.6000	296798.4602	539297.7000	352347.8244	0.173794	0.236914	-0.4497	False
macro_contrast_std	0.3815	0.3704	0.5442	0.3703	0.183811	0.236914	-0.4392	False
medium_homogeneity_d3	0.9401	0.0441	0.9189	0.0521	0.183839	0.236914	0.4392	False
region_std_intensity	12.7365	2.0479	11.9472	1.5191	0.185234	0.236914	0.4378	False
medium_correlation_d5	0.8844	0.0152	0.8896	0.0078	0.187090	0.236914	-0.4359	False
medium_homogeneity_mean	0.9361	0.0468	0.9141	0.0548	0.191482	0.236914	0.4315	False
macro_homogeneity_d7	0.9285	0.0519	0.9046	0.0600	0.196794	0.236914	0.4262	False
macro_homogeneity_mean	0.9285	0.0519	0.9046	0.0600	0.196794	0.236914	0.4262	False
medium_homogeneity_d5	0.9322	0.0495	0.9094	0.0574	0.198590	0.236914	0.4245	False
macro_correlation_d7	0.8518	0.0197	0.8575	0.0103	0.266660	0.307337	-0.3658	False
macro_correlation_mean	0.8518	0.0197	0.8575	0.0103	0.266660	0.307337	-0.3658	False
lbp_hist_4	0.0238	0.0014	0.0243	0.0017	0.271544	0.307750	-0.3620	False
medium_homogeneity_std	0.0041	0.0028	0.0049	0.0028	0.344780	0.384344	-0.3104	False
medium_correlation_std	0.0256	0.0033	0.0248	0.0018	0.360392	0.395269	0.3004	False
macro_dissimilarity_std	0.0106	0.0087	0.0125	0.0094	0.507106	0.547352	-0.2173	False
macro_correlation_std	0.0018	0.0019	0.0022	0.0021	0.526871	0.559801	-0.2072	False
lbp_hist_3	0.0283	0.0012	0.0285	0.0014	0.608009	0.628069	-0.1678	False
lbp_hist_2	0.0262	0.0018	0.0260	0.0013	0.609596	0.628069	0.1671	False
region_kurtosis	0.1481	0.2630	0.1032	0.3693	0.668796	0.678778	0.1399	False
region_mean_intensity	99.1980	13.8883	99.0903	11.3831	0.979278	0.979278	0.0085	False

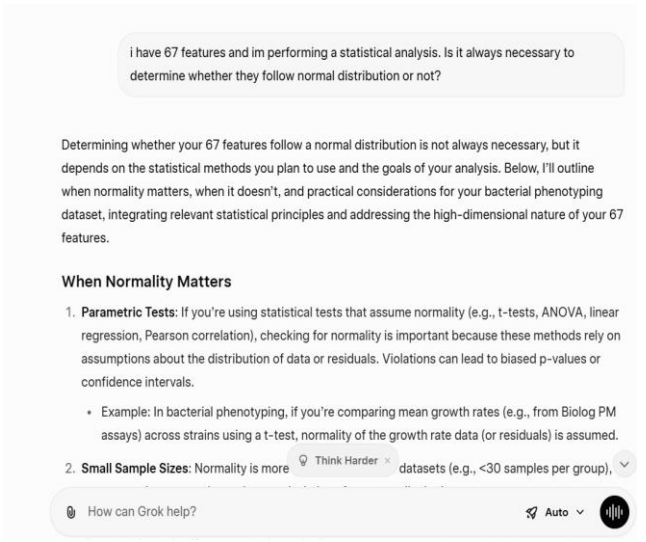
Appendix F

F1. AI Usage









Bibliography

Alva, A. et al., 2013. Morphological Characterization of Mycobacterium tuberculosis in a MODS Culture for an Automatic Diagnostics through Pattern Recognition. *PLoS ONE*, 16 12, 8(12), p. e82809.

Armi, L. & Fekri-Ershad, S., 2019. Texture image analysis and texture classification methods - A review. *arXiv.org*.

Benjamini, Y. & Hochberg, Y., 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1), pp. 289-300.

Boehme, C. et al., 2010. Rapid Molecular Detection of Tuberculosis and Rifampin Resistance. *New England Journal of Medicine*, pp. 1005-1015.

Caviedes, L. et al., 2000. Rapid, Efficient Detection and Drug Susceptibility Testing of Mycobacterium tuberculosis in Sputum by Microscopic Observation of Broth Cultures. *Journal of Clinical Microbiology*, 3, 38(3), pp. 1203-1208.

Comina, C. et al., 2011. Development of an automated MODS plate reader to detect early growth of Mycobacterium tuberculosis. *Journal of Microscopy*, 20 1, 242(3), pp. 325-330.

Dheepak, G., 2024. Optimizing Brain Tumor Classification through Innovative Texture-Based MRI Analysis: A Fusion of Novel GLCM and LBP with SVM Ensemble for Enhanced Accuracy. *2024 Global Conference on Communications and Information Technologies (GCCIT)*, pp. 1-6.

Gagniuc, E., 2025. Image Preprocessing Techniques. In: *Computational Analysis of Microscope Images*. s.l.:Springer, Cham, pp. 27-64.

Ghanegolmohammadi, F., Eslami, M. & Ohya, Y., 2024. Systematic data analysis pipeline for quantitative morphological cell phenotyping. *Computational and Structural Biotechnology Journal*, 1 12, Volume 23, pp. 2949-2962.

Guindon, B. & Zhang, Y., 2016. Application of the Dice Coefficient to Accuracy Assessment of Object-Based Image Classification. *Canadian Journal of Remote Sensing*, 43(1), pp. 48-61.

Haralick, R., 1973. Textural Features for Image Classification. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-3(6), pp. 610-621.

Hlabangana, S., 2024. *Complex cords: Automated Quatitative Analysis of Mycobacterial Cording Phenotypes*, Cape Town: Universty of Cape Town.

Jihadah, M. N. et al., 2025. The accuracy of Mycobacterium tuberculosis complex identification using cording observation with mpt64 antigen as the reference standard. *Diagnostic Microbiology and Infectious Disease*, 29 4, 113(2), p. 116927.

Julián, E. et al., 2010. Microscopic Cords, a Virulence-Related Characteristic of Mycobacterium tuberculosis, Are Also Present in Nonpathogenic Mycobacteria. *Journal of Bacteriology*, 1 4, 192(7), p. 1751–1760.

Julián, E. et al., 2010. Microscopic Cords, a Virulence-Related Characteristic of Mycobacterium tuberculosis, Are Also Present in Nonpathogenic Mycobacteria. *Journal of Bacteriology*, 192(7), p. 1751–1760.

Kaslum, S. et al., 2021. A high-throughput screening assay based on automated microscopy for monitoring antibiotic susceptibility of Mycobacterium tuberculosis phenotypes. *BMC Microbiology*, Volume 21.

Lopez-Garnier, S., Sheen, P., Zimic, M. & Olier, I., 2019. Automatic diagnostics of tuberculosis using convolutional neural networks analysis of MODS digital images. *PLOS ONE*, 27 2, 14(2), p. e0212094.

Mann, H. & Whitney, D., 1947. On a Test of Whether One of Two Random Variables Is Stochastically Larger than the Other. *The Annals of Mathematical Statistics*, 18(1), pp. 50-60.

McNerney, R. et al., 2012. Tuberculosis Diagnostics and Biomarkers: Needs, Challenges, Recent Advances, and Opportunities. *The Journal of Infectious Diseases*, 205(suppl_2), pp. S147-S158a.

Middlebrook, G., Dubos, R. & Pierce, C. H., 1947. VIRULENCE AND MORPHOLOGICAL CHARACTERISTICS OF MAMMALIAN TUBERCLE BACILLI. *Journal of Experimental Medicine*, 1 8, 86(2), pp. 175-184.

Mishra, R. et al., 2023. Mechanopathology of biofilm-like Mycobacterium tuberculosis cords. *Cell*, 186(23), pp. 5135-5130.

Mizrahi, V., 2024. A Spotlight on the Tuberculosis Epidemic in South Africa. *Nature Communications*, 12 2, 15(1), p. 1290.

Moore, D. et al., 2006. Microscopic-Observation Drug-Susceptibility Assay for the Diagnosis of TB. *New England Journal of Medicine*, 355(15), pp. 1539-1550.

Núñez-Fernández, D. et al., 2020. Automatic semantic segmentation for prediction of tuberculosis using lens-free microscopy images. *arXiv (Cornell University)*, 1 1.

Organization, W. H., 22. *Global Tuberculosis Report 2022.* [Online] Available at: https://www.who.int/publications/i/item/9789240061729

Organiztion, W. H., 2022. *World Health Organization.* [Online].

Rakesh, K. et al., 2018. A Review of Medical Image Segmentation Algorithms. *EAI Endorsed Transactions on Pervasive Health and Technology*, 7(27), p. 169184.

Rodriguez, J. et al., 2022. Evaluation of three alternatives cost-effective culture media for Mycobacterium tuberculosis detection and drug susceptibility determination using the microscopic observation drug susceptibility (MODS) assay. *Tuberculosis*, Volume 137, p. 102273.

Shapiro, S. & Wilk, M., 1965. An Analysis of Variance Test for Normality (Complete Samples). *Biometrika*, 52(3/4), pp. 591-611.

Solis, L. et al., 2016. Evaluation of a lens-free imager to facilitate tuberculosis diagnostics in MODS. *Tuberculosis*, Volume 97, pp. 26-32.

Steingart, K. et al., 2006. Fluorescence versus conventional sputum smear microscopy for tuberculosis: a systematic review. *The Lancet Infectious Diseases*, 9, 6(9), pp. 570-581.

Swarts, B. et al., 2012. Probing the Mycobacterial Trehalome with Bioorthogonal Chemistry. *Journal of the American Chemical Society*, 24 9, 134(39), pp. 16123-16126.

Vyas, R., 2017. *Co-occurrence features and neural network classification approach for iris recognition.* [Online] Available at: https://www.researchgate.net/figure/GLCM-matrix-formation-Different-directions-for-calculating-the-frequencies-of-various_fig1_323718838
[Accessed 06 10 2025].

Zayed, N., 2015. Statistical Analysis of Haralick Texture Features to Discriminate Lung Abnormalities. *International Journal of Biomedical Imaging*, 2015(1), pp. 1-7.